

PEAK3

PRODUCT + IMPLEMENTATION BLUEPRINT

A public NBA peak index,
transparent formula explorer,
and globally ranked
competitive drafting game.



INDEX Explore every peak

ARENA Draft and compete

LAB Understand the model

BUILD BLUEPRINT

A focused plan for turning PEAK3 into a public basketball research platform and a globally ranked competitive game.

Version 1.0 | June 27, 2026 | Product, game systems, design, data, engineering, launch, and operations

DOCUMENT PURPOSE

This is an implementation plan, not an idea inventory. It identifies the product promise, launch scope, game rules, ranking systems, information architecture, visual language, technical boundaries, data contracts, safety controls, roadmap, and acceptance gates required to begin building.

The single-sentence product

Draft exact NBA peaks, compete on fair shared boards, and understand every result through a transparent model.

Proposed PEAK3 product promise

What this plan recommends

- **One umbrella, three surfaces.** PEAK3 INDEX is the public statistical reference, PEAK3 ARENA is the game, and PEAK3 LAB explains and exposes the model.
- **One flagship loop.** Five-round exact-peak drafting with live Lineup DNA, transparent scoring receipts, global rankings, daily boards, and challenge links.
- **Three competitive identities.** Separate 1Y, 3Y, and 5Y ladders, plus a Triple Crown that rewards mastery across all three.
- **Two progression systems.** Glicko-2 measures skill; XP, streaks, collections, and achievements reward participation without corrupting matchmaking.
- **One trust contract.** Every official score is reproducible, versioned, auditable, and explainable. Experimental lineup logic is visibly separated from the validated PEAK3 individual model.
- **A disciplined launch.** Ship a polished core game and credible index first. Do not dilute the first release with auctions, live rooms, era simulations, or dozens of lifelines.

North-star outcomes

Outcome	What success looks like	Primary measure
Instant comprehension	A new visitor understands the game and makes a first selection without reading a long tutorial.	First-card selection rate; time to first choice
Repeat play	Players return to beat their score, climb the ladder, protect a streak, or answer a challenge.	D1/D7/D30 retention; games per active player
Competitive legitimacy	Strong players rise through fair, auditable competition rather than raw grinding.	Rating calibration; rematch rate; score dispute rate
Research authority	Analysts and creators use PEAK3 pages, exports, and citations as a common reference.	Organic links; exports; API usage; cited model versions

Outcome	What success looks like	Primary measure
Share-driven growth	A shared result is playable and informative, not merely a screenshot.	Challenge starts per share; completion conversion
Healthy performance	The phone experience remains fast and stable despite rich visuals.	Core Web Vitals; error-free sessions; rage-click rate

Table 1. North-star outcomes and instrumentation.

CONTENTS

Implementation map

Part	Topic
01	Vision, strategic thesis, and boundaries
02	Competitive landscape and lessons
03	Audience, jobs, and product principles
04	Brand architecture and information architecture
05	Flagship Peak Draft specification
06	Competitive modes and global ranked system
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10	PEAK3 Index, player pages, comparisons, and public data
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16	Roadmap, team plan, and execution cadence
17	Launch, creators, community, and growth
18	Scope priorities, acceptance criteria, and open decisions
A	Appendices: schemas, events, resources, and checklists

THE PEAK3 PRODUCT FLYWHEEL

Credibility powers play. Play powers distribution. Distribution powers the data ecosystem.

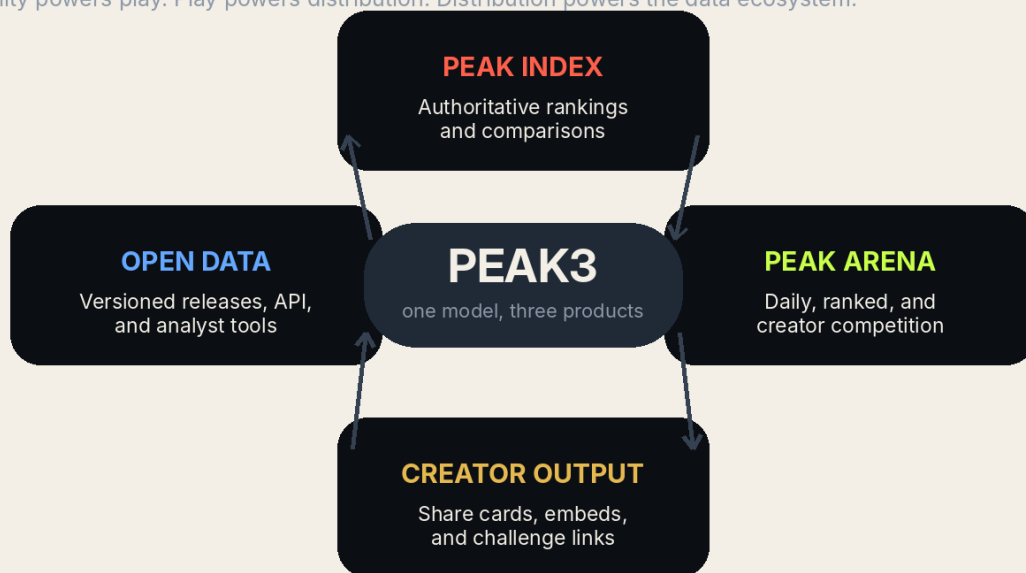


Figure 1. The product flywheel: credible model → meaningful play → social competition → richer data and debate → greater model authority.

HOW TO USE THIS BLUEPRINT

Read Parts 1-4 to align on the product. Treat Parts 5-10 as the product requirements document. Parts 11-15 are the implementation and quality contract. Parts 16-18 define the build sequence and shipping gates.

PART 01

Vision, strategic thesis, and boundaries

The product should become the place basketball fans research a peak, debate a peak, draft a peak, and prove what they know.

The opportunity

Basketball fans already spend enormous time ranking individual seasons, comparing primes, constructing hypothetical lineups, and arguing about how eras should be evaluated. Existing viral games prove that a five-minute team-building loop can spread quickly. Existing analytics games prove that advanced metrics can support deeper strategy. Neither category fully combines a defensible peak model, accessible explanation, satisfying lineup construction, fair competition, and a reusable public data platform.

PEAK3 already owns the hardest intellectual asset: a deterministic framework that separates regular-season impact, two-way production, recognition, postseason performance, team success, and contextual adjustments across exact seasons and contiguous windows. The website should not hide that work behind a single novelty score. It should turn that work into a public institution and a game system.

Strategic thesis

The index earns trust. The game earns reach. The formula explorer converts skepticism into curiosity. The shared board converts curiosity into competition.

PEAK3 growth thesis

- **Credibility is the moat.** The game can be copied more easily than the model, methodology, provenance, version history, and explanation system.
- **Exact peaks are the content primitive.** Every card, comparison, leaderboard, result, and story should reference a specific season or a contiguous 3Y/5Y window.
- **Explainability is part of the entertainment.** Users should inspect why a result happened, challenge assumptions, and share the receipt.
- **Fair boards create real skill.** Ranked competition must compare users on the same offers, order, rules, model version, and tool budget.
- **The best social object is playable.** A share should launch the same challenge, not stop at an image.
- **Depth should unfold progressively.** The first choice is easy; the detailed model is available one click deeper, not forced into the first minute.

Product architecture

Surface	Purpose	Core visitor question
PEAK3 INDEX	Canonical leaderboards, player histories, season pages, comparisons, and model-version records.	Who peaked highest, and what does the evidence say?
PEAK3 ARENA	Daily, ranked, challenge, and tournament drafting built around exact peaks.	Can I construct a better lineup than everyone else?
PEAK3 LAB	Interactive formula explorer, data browser, exports, methodology, and developer resources.	How was this calculated, and can I analyze it myself?

Table 3. Three surfaces, one shared data and identity system.

Non-negotiable boundaries

- ❑ Do not market an experimental hypothetical-lineup layer as though PEAK3 has scientifically predicted an 82-game record.

- ❑ Do not collapse individual peak quality, roster coverage, and player fit into an unexplained score.
- ❑ Do not let paid features affect ranked outcomes, board quality, tools, retries, or matchmaking.
- ❑ Do not require sign-in before a visitor can complete a first unranked game.
- ❑ Do not launch with more modes than the interface can explain in one short mode selector.
- ❑ Do not silently change old game results when weights, data, or lineup logic changes.
- ❑ Do not let streaks punish users for living in the wrong timezone or missing a day for real life.
- ❑ Do not use NBA/team logos, player photos, or third-party visual assets without a clear rights strategy.
- ❑ Do not make the desktop concept beautiful at the expense of a fast, legible phone experience.

What “generational” means in practical terms

A generational product is not defined by feature count. It creates a recognizable object, vocabulary, and ritual. For PEAK3, the object is the exact peak card, the vocabulary is 1Y/3Y/5Y plus Lineup DNA and score receipts, and the ritual is the daily shared board and Triple Crown. The goal is for fans to say “What is his PEAK3?” or “Send me your board” in the same effortless way they reference an all-time list or a fantasy matchup.

PART 02

Competitive landscape and lessons

Borrow the retention mechanics that work, not the identity or surface treatment of another game.

82-0: viral clarity, weak evaluation contract

Strength to borrow	Weakness to avoid	PEAK3 response
The game is explainable in one sentence.	A final record can imply false precision.	Use a Peak Rating, percentile, division movement, and transparent receipts.
Random constraints create tension and regret.	Independent random rolls undermine direct competitive fairness.	Use daily/shared seeded boards for ranked play; reserve chaos for casual play.
A short game produces shareable lineups.	Repeated screenshots become undifferentiated feed spam.	Every result contains a playable challenge and one distinctive analytical insight.
Famous player decisions create immediate recognition.	Bland box-score aggregation overvalues volume and ignores context or fit.	Use exact PEAK3 windows plus a bounded, published lineup layer.

Table 4. How PEAK3 should learn from the 82-0 loop without cloning it.

SixRings: strategic depth, metric dependence

Strength to borrow	Risk to avoid	PEAK3 response
Exact player-seasons and blind information reward knowledge.	The metric can become the game; veterans memorize favored cards.	Rotate 1Y/3Y/5Y, constraints, archetypes, and explanation-led decisions.
Shared boards and competitive modes support fair comparison.	Mode and lifeline proliferation can overwhelm the simple core.	Launch with two tools and three clearly named peak windows.
Lineup bonuses recognize that five stars are not automatically a team.	Discrete bonuses can feel bolted on or arbitrary.	Expose continuously updating Lineup DNA and a bounded fit receipt.
Trophies, Elo, and tournaments create recurrence.	A single public score may blend skill and play volume.	Separate Glicko-2 skill, seasonal points, XP, and record books.

Table 5. How PEAK3 should learn from SixRings while keeping its own identity.

Adjacent systems worth adapting

Reference pattern	Why it works	Adaptation for PEAK3
Chess ratings	Opponent-aware ratings estimate skill rather than total activity.	Glicko-2 per queue with uncertainty, placements, provisional status, and decay rules.
GeoGuessr ranked divisions	Visible tiers translate a numeric rating into identity and goals.	Prospect through Legend divisions with seasonal crests and promotion moments.
Strava segments	Global, friends, age/region filters, personal records, and recurring leaderboards create multiple attainable competitions.	Global, friends, regional opt-in, creator, daily-board, and personal-best views.
Duolingo streak protection	A streak can create a return ritual when it includes recovery and forgiveness.	One weekly reserve token, local-day handling, and no rating advantage.
Daily puzzle games	One shared artifact makes conversation comparable and spoiler-safe.	Daily Peak and Sunday Triple Crown boards with stable seed and model version.
Fantasy drafts	Budgets, scarcity, roster slots, and role tradeoffs create enduring strategy.	Later: auction and salary-cap formats, only after the core draft proves retention.

Table 6. External product patterns that fit PEAK3's competitive goals.

Distinctive position

**PEAK3 = research authority + exact peaks + fair drafting + transparent lineup logic
+ playable sharing**

The product should never be introduced as a “better 82-0.” That framing makes the incumbent the category owner. The public description should emphasize exact peaks, transparent receipts, and ranked shared-board drafting. Competitors are useful validation, not the brand vocabulary.

PART 03

Audience, jobs, and product principles

Design one accessible core for casual fans, then reveal analytical depth for the people who seek it.

Primary audiences

Audience	Job to be done	First value moment	Long-term reason to return
Competitive NBA fan	Prove basketball knowledge and beat friends or the world.	Completes a board and sees a percentile/rank movement.	Seasonal ladder, Triple Crown, rivals, personal records.
Analytics-minded fan	Interrogate rankings and understand why seasons differ.	Expands the score receipt or formula layer.	Comparisons, Lab queries, model updates, datasets.
Content creator	Generate arguments, challenges, visuals, and audience participation.	Publishes a creator board with a leaderboard.	Recurring series, audience analytics, export templates.
Statistician / developer	Use clean, documented, versioned data.	Downloads a release or queries an API endpoint.	Stable IDs, changelog, provenance, reproducible analyses.
New/casual fan	Learn famous seasons without feeling excluded.	Gets useful card cues and learns from the result explanation.	Discovery mode, streaks, collections, approachable comparisons.

Table 7. Primary audiences and their value moments.

Product principles

1. Immediate action before explanation. A visitor should see a card decision above the fold and reach a satisfying result before being asked to learn the full formula.
2. Depth on demand. Every score, component, card, and chart should reveal deeper layers without turning the default screen into a research paper.
3. Competitive truth. A ranked result is valid only when both competitors faced equivalent information and server-verifiable rules.
4. Model humility. PEAK3 should state assumptions and uncertainty rather than hiding behind false precision or branding every output as objective truth.
5. Skill and habit are different. Rating measures competitive performance; XP and streaks measure activity. The UI and reward system must never confuse them.
6. Mobile is the primary game board. Desktop can expose more analysis, but no decision, card, result, or leaderboard should require desktop width.
7. Motion explains state. Animation should clarify dealing, selection, score composition, promotion, and hierarchy; never block play or create visual fatigue.
8. Every public output is versioned. A ranking or game should remain reproducible after the model evolves.
9. Accessible by default. Keyboard operation, visible focus, reduced motion, color-independent encoding, readable contrast, and screen-reader labels are launch requirements.
10. Community debate without community degradation. Build reporting, rate limits, display-name controls, and spoiler handling before large-scale social surfaces.

Core user journeys

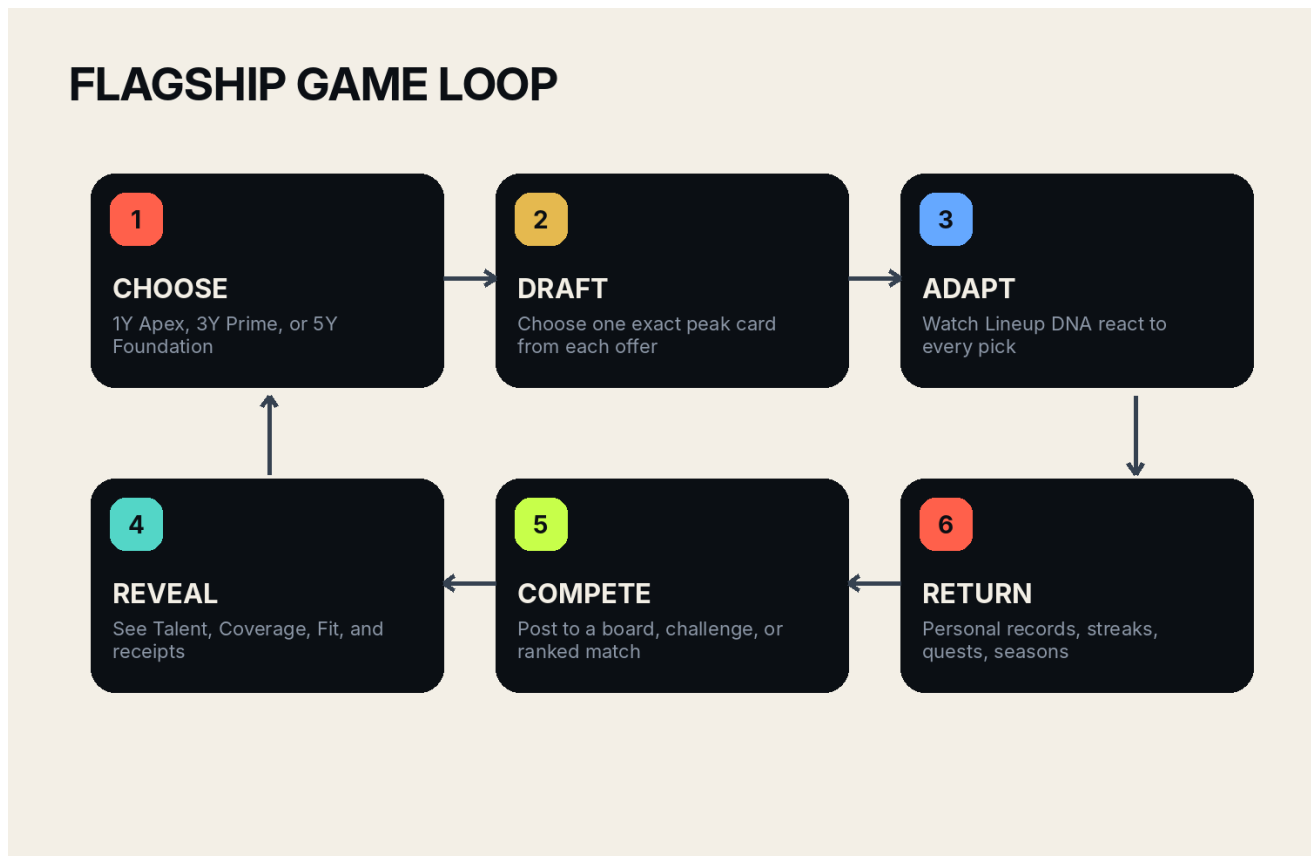


Figure 2. The first-session game loop: land → choose mode → draft → understand result → challenge or join ranked → return.

Anonymous first session

1. Open the home page and immediately see Today's Peak Draft.
2. Select 1Y Apex, receive a two-line rules overlay, and make five choices.
3. See Peak Rating, global-board percentile, Lineup DNA, and three score receipts.
4. Create a challenge link or sign in to save the result, claim XP, and enter placements.

Returning ranked player

1. Land on a personalized Arena home showing current division, today's board, active challenges, and one recommended queue.
2. Play an asynchronous same-board matchup or daily ranked board.
3. Receive rating movement only after a valid comparative result exists.
4. Review one decisive choice, personal-best progress, rival activity, and next milestone.

Research journey

1. Search for a player and open the career Peak Mountain.
2. Switch between 1Y, 3Y, and 5Y windows.
3. Compare exact windows and inspect component differences.
4. Open the formula explorer at the highlighted component or export the rows with model-version metadata.

PART 04

Brand architecture and information architecture

A clear global shell lets the game feel immediate while the Index and Lab retain institutional depth.

Recommended naming system

Name	Role	Public-language example
PEAK3	Umbrella brand and model.	"Powered by PEAK3 v1.2."
PEAK3 ARENA	Competitive game surface.	"Play Today's Apex board."
PEAK3 INDEX	Leaderboards, profiles, seasons, and comparisons.	"Explore the highest 3Y peaks."
PEAK3 LAB	Formula explorer, data tools, methodology, and API.	"Trace exactly how this score was built."
Peak Draft	Flagship five-round drafting format.	"Challenge me on this Peak Draft."
Triple Crown	Three-board 1Y/3Y/5Y competitive set.	"Win across every kind of peak."
Lineup DNA	Live roster capability visualization.	"Your lineup lacks secondary creation."
Peak Receipt	Explainable final-score breakdown.	"See what added and cost points."

Table 8. Recommended product vocabulary.

Global navigation

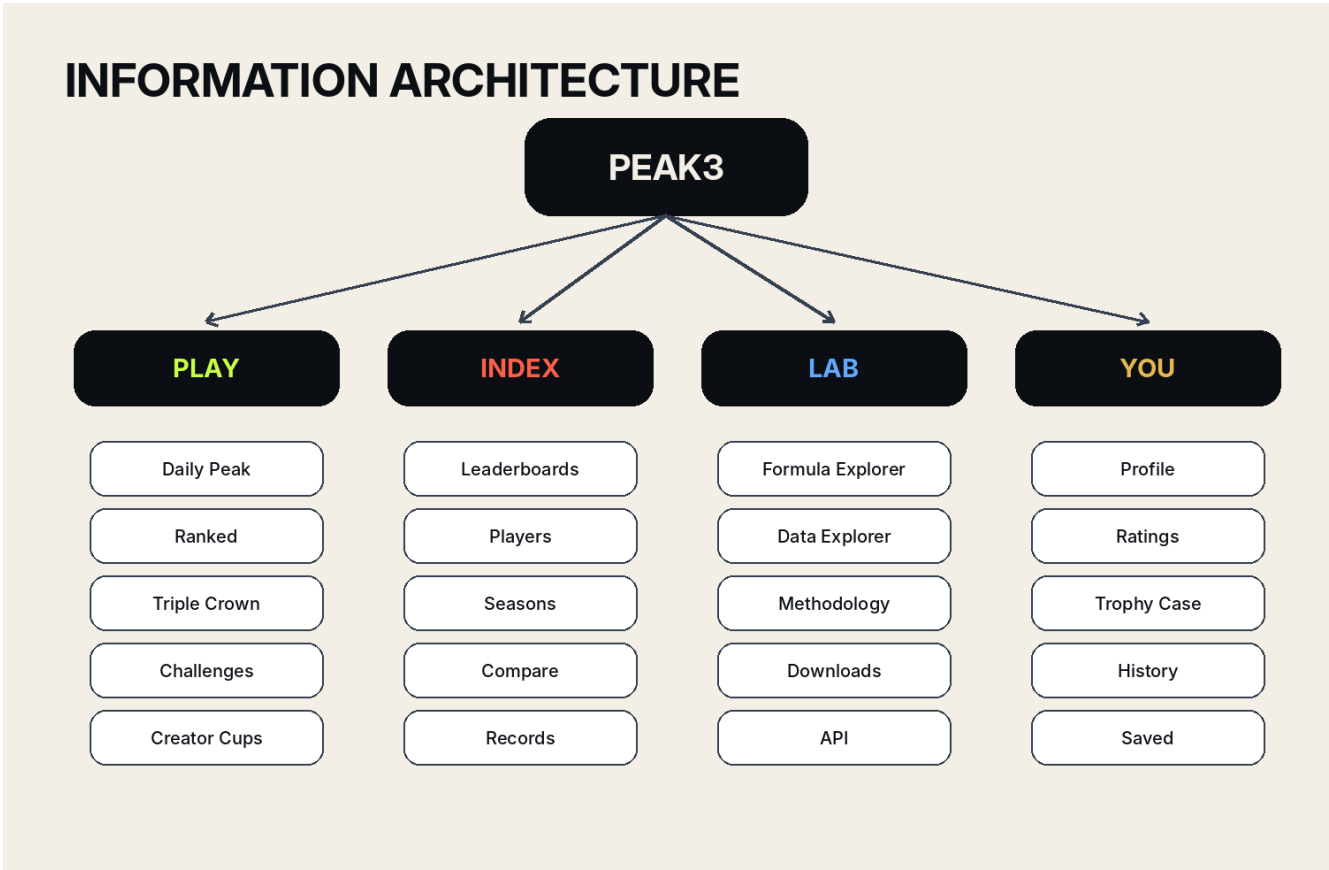


Figure 3. Information architecture: a common identity, search, notification, and model-version layer across Index, Arena, and Lab.

Desktop navigation	Mobile destination	Required contents
Play	Arena tab	Today, Ranked, Challenges, Triple Crown, Practice, Match history
Index	Index tab	Leaderboards, Players, Seasons, Compare, Records
Lab	Lab tab	Formula, Data Explorer, Downloads, API, Methodology, Changelog
Search	Persistent search action	Players, exact seasons, peak windows, game boards, creators
Profile	Profile tab	Division, ratings, XP, achievements, trophy case, saved comparisons, privacy

Table 9. Global navigation contract.

Home page composition

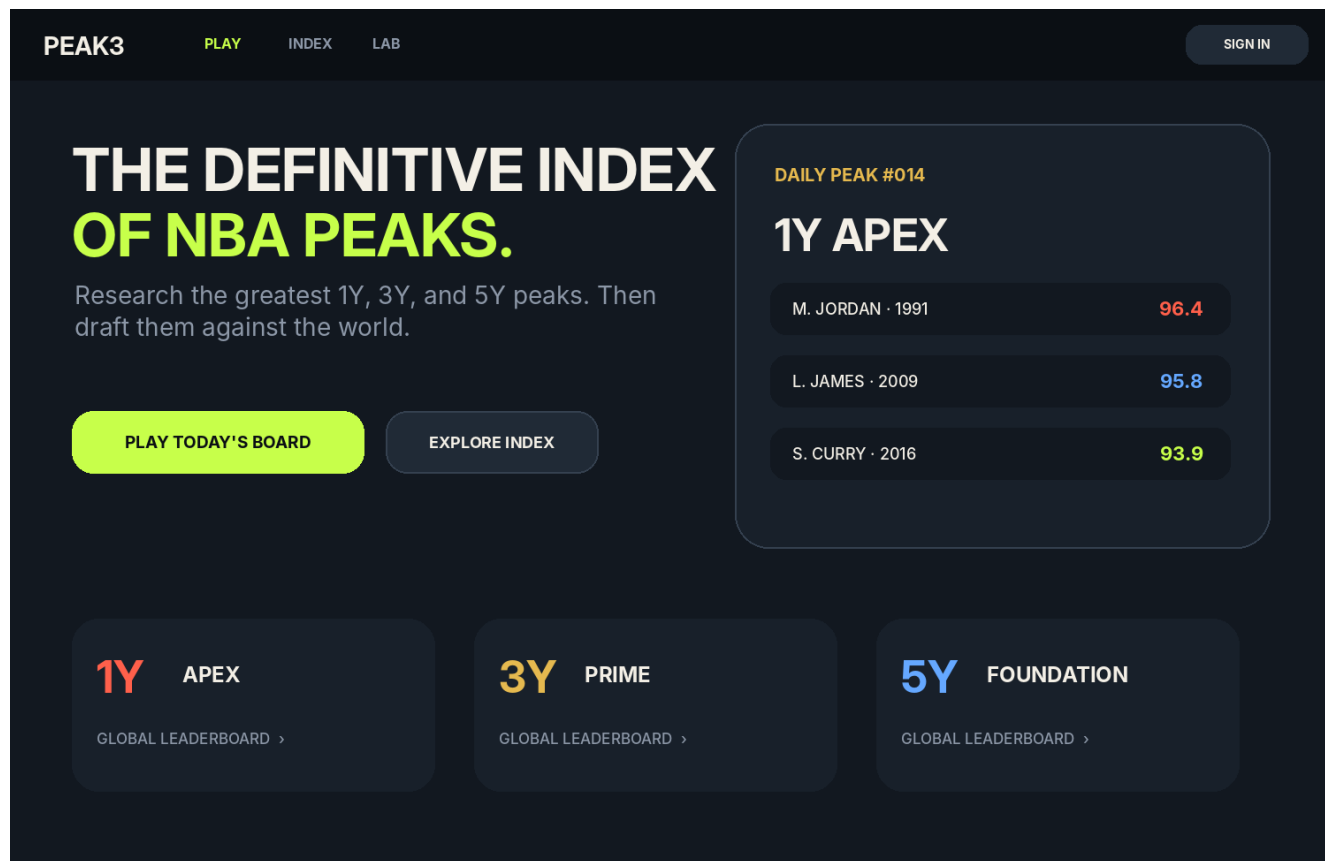


Figure 4. Early home-page direction: one decisive playable surface, immediate credibility, and progressive analytical depth.

1. Hero: Today's Peak Draft with a real first decision or a single Play Now action; one-line value proposition; no carousel.
2. Proof strip: current model version, seasons covered, player-seasons evaluated, and a direct methodology link.
3. Three peak leaders: top 1Y, 3Y, and 5Y windows with a swipeable mobile presentation.
4. Live competition: current global #1, friends activity, and the player's own rank or anonymous percentile.
5. Interactive teaser: one "Who Peaked Higher?" comparison that reveals a formula receipt.
6. Research surface: trending comparisons, newly updated data, and a creator/community feature.
7. Footer: methodology, sources, legal, accessibility, status, developer resources, and model changelog.

Page-level URL conventions

Resource	Canonical pattern	Notes
Player	/index/players/{player-slug}	Stable internal ID, slug can redirect after changes.

Resource	Canonical pattern	Notes
Exact season	/index/players/{player-slug}/{season}	Season label uses ending year convention consistently.
Peak window	/index/windows/{window-id}	Immutable ID includes model-version relation.
Comparison	/compare?windows={id1},{id2}	Shareable and reproducible; encode model version.
Daily board	/arena/daily/{yyyy-mm-dd}/{mode}	Same seed for all users; spoiler-safe metadata.
Challenge	/c/{public-token}	Token resolves to immutable board/rules snapshot.
Ranked match	/arena/matches/{public-id}	Public view redacts private anti-cheat fields.
Formula node	/lab/formula/{node-id}?version={version}	Deep-link to exact component and version.
Data release	/lab/releases/{semver}	Persistent page with files, schema, provenance, and changelog.

Table 10. Stable, shareable URL structure.

Responsive behavior

- **Phone (< 640 px).** One decision at a time, card stack or horizontal carousel, persistent lineup tray, bottom navigation, thumb-reachable primary action, compact DNA visualization.
- **Tablet.** Two-pane game layout with choices and lineup; formula explorer uses zoomable canvas plus bottom sheet.
- **Desktop.** Three-card comparison, full lineup rail, contextual receipts, keyboard shortcuts, multi-column Index analysis.
- **Wide desktop.** Never stretch body copy indefinitely. Use additional room for persistent comparison panels, not larger line lengths.

PART 05

Flagship Peak Draft specification

The core game must be simple enough to learn in one round and deep enough to remain interesting after hundreds of rounds.

Core rule set

ONE-SENTENCE RULES

Choose one exact peak from three offers in each of five rounds. Fill five functional lineup roles, use two strategic tools, and finish with the strongest explainable lineup you can build.

Element	Launch rule	Reason
Rounds	5	One decision per roster slot; short enough for repeat play.
Offers per round	3 exact peak cards	Creates comparison and regret without analysis paralysis.
Peak windows	1Y, 3Y, or 5Y depending on queue	The central PEAK3 distinction becomes the competitive format.
Roles	Lead Creator; Guard/Wing; Wing/Forward; Forward/Big; Anchor	Functional roles travel better across eras than rigid traditional positions.
Tools	One Reframe and one Hold	Meaningful agency with a tiny learning burden.
Timer	None in Daily/Practice; optional 45-60 seconds per choice in higher ranked tiers	Keep onboarding calm; add time pressure where competitive integrity requires it.
Result	Peak Rating + board percentile + Lineup DNA + receipts	Avoid fake win totals; reveal why the result happened.

Table 11. Launch rules for Peak Draft.

Card anatomy

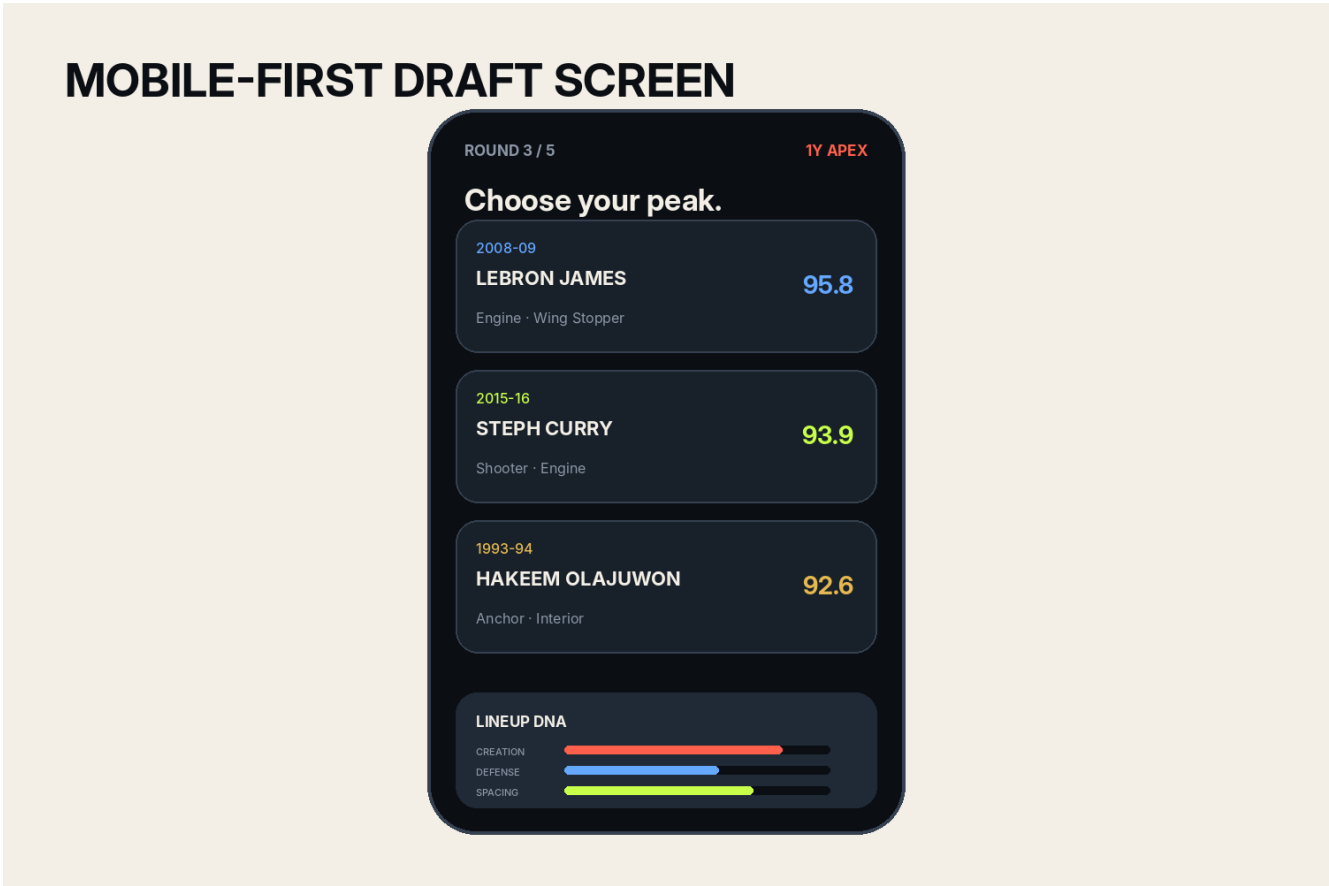


Figure 5. Draft-screen concept: exact peak cards, role eligibility, live Lineup DNA, tool state, and minimal distraction.

Always visible	Revealed on tap / hover	Hidden or mode-dependent
Player name; exact season/window; role eligibility; team text; PEAK3 score band; archetype tags.	Component fingerprint; key stats; postseason context; score methodology; data completeness.	Exact total score in Blind Resume; player identity in identity-hidden modes; detailed stats in HooplQ-style variants.

Table 12. Progressive disclosure on a peak card.

Functional roles

Role	Minimum interpretation	Common qualifying profiles
Lead Creator	Can drive half-court offense as the primary initiator.	High-volume creators, heliocentric wings, elite playmaking guards.
Guard / Wing	Can defend or operate on the perimeter without requiring primary control.	Two-way guards, off-ball shooters, secondary creators, point-of-attack defenders.
Wing / Forward	Provides scalable size, defense, shooting, creation, or connective play.	Big wings, combo forwards, versatile two-way stars.
Forward / Big	Can survive frontcourt matchups and contribute size-dependent value.	Power forwards, mobile centers, jumbo wings with credible interior function.
Anchor	Provides the roster’s most important interior defensive, rebounding, or size role.	Centers, rim protectors, dominant rebounders, switchable defensive bigs.

Table 13. Launch role taxonomy. Eligibility should be documented and editable through a reviewed data layer.

Strategic tools

Reframe

Replace all three current offers once per game. Reframe is powerful when a board has a severe role mismatch, but it sacrifices the known options. The server records the discarded offer set and regenerated deterministic branch so the move can be replayed and verified.

Hold

Place one unselected card into a temporary slot. The next round contains the held card plus two new offers. Selecting another card releases the held card; selecting the held card consumes the slot. Hold creates planning, opportunity cost, and memorable regret without adding a complicated inventory.

DO NOT ADD MORE AT LAUNCH

Spy, swap, duplicate, reveal, position filters, flash effects, and multiple rerolls may be fun later, but each one increases balance work, tutorial burden, anti-cheat surface area, and mobile clutter. Launch depth should come from cards, roles, windows, and shared boards.

Lineup DNA

Lineup DNA is the live strategic feedback system. It is not the final score. Each selection updates a compact roster profile and highlights one emerging strength or weakness. The system should avoid declaring a “correct” choice before the draft ends; it should expose tradeoffs.

Dimension	Question it answers	Example feedback
Primary creation	Can someone reliably create efficient offense against set defenses?	“Elite engine established.”
Secondary playmaking	Can the offense survive traps, switches, and off-ball possessions?	“One more connector would stabilize creation.”
Scoring pressure	Does the lineup bend defenses at the rim, line, midrange, or through volume?	“High pressure, but concentrated in one player.”
Spacing / off-ball value	Can stars coexist without every player controlling the ball?	“Two non-shooting frontcourt roles now overlap.”
Perimeter defense	Can the lineup defend elite guards and wings?	“Point-of-attack weakness remains.”
Interior defense	Can it protect the rim and survive frontcourt size?	“Anchor role is filled, but switching is limited.”
Rebounding / possession control	Can it end defensive possessions and create second chances?	“Below-board median on defensive rebounding.”
Availability / postseason translation	Did the selected peaks sustain value under playoff and availability demands?	“High ceiling; moderate durability risk in this window.”

Table 14. Lineup DNA dimensions and user-facing language.

Game-state machine

State	Server/client behavior	Exit condition
board_loaded	Immutable board seed, mode, model version, card pool version, ruleset hash, and offer tree are loaded.	Client acknowledges valid snapshot.
round_active	Current offers and eligible tools are displayed; client starts local interaction telemetry.	Valid card placement, tool use, abandonment, or timeout.
placement_pending	Client proposes card/role; server validates eligibility and previous state hash.	Server returns authoritative next state.

State	Server/client behavior	Exit condition
draft_complete	Server computes official lineup result and board-relative metrics.	Result snapshot persists.
challenge_ready	Share token and spoiler-safe image payload are created.	Share, rematch, sign-in, or navigation.
rated_pending	Ranked submission waits for comparable opponent or board settlement.	Rating job completes or match is invalidated.

Table 15. Authoritative state machine for replayability and anti-cheat.

Result screen



Figure 6. Result direction: status and story first, formula receipts second, replay/challenge actions always visible.

1. Outcome line: Peak Rating and percentile, plus win/loss or rating movement when the match is settled.
2. Lineup: five exact cards with roles and one-tap access to each window.
3. Lineup DNA: final capability profile compared with the board median and the player's recent average.
4. Peak Receipt: top positive interaction, largest weakness, most valuable pick, and most consequential alternative.
5. Decision replay: round-by-round offers, selected card, held/reframed branches, and model delta estimates.
6. Progress: rating, division, XP, streak, personal best, achievement, or record changes; never overload with all at once.
7. Actions: challenge a friend, play next queue, review formula, save/share, compare selected peaks.

Board generation rules

- **Deterministic seed.** Every official board resolves from a signed seed, pool version, mode, and ruleset hash.
- **Controlled difficulty.** Boards are sampled to create real tradeoffs and viable role completion, not five trivial superstar rounds or unavoidable dead ends.
- **Diversity constraints.** Cap repeated players, eras, franchises, and obvious top-tier cards; prevent duplicate exact windows in one board.
- **Role feasibility.** At least one completion path exists without requiring Reframe. Ranked boards are simulation-checked before publication.

- **Information balance.** Cards shown together should not always differ by obvious total-score tiers; include archetypal tradeoffs.
- **Retrospective audit.** Store generated candidates, final offer graph, simulation distribution, and publication checks for disputes and tuning.

PART 06

Competitive modes and global ranked system

Competition should create a durable identity, not a leaderboard that merely rewards whoever plays the most.

Mode portfolio

Mode	Core format	Rating impact	Launch priority
Daily Peak	One shared board per mode per local/global day; score against all finishers.	Optional daily-rated queue after placements; always has board percentile.	P0
Async Ranked Duel	Two users play the same hidden board independently.	Yes: Glicko-2 after both complete and integrity checks pass.	P0
Direct Challenge	A player sends an immutable board to a friend or public audience.	No by default; ranked challenges require queue rules.	P0
Triple Crown	One 1Y, one 3Y, and one 5Y board; combined normalized performance.	Separate Crown rating or event score after launch.	P1
Practice	Unrated boards with hints, formula previews, and unlimited replay variants.	No.	P0
Creator Gauntlet	Creator publishes a board set and branded public leaderboard.	No global rating; event standings only.	P1
Constraint Draft	Shared special rules: salary cap, decade limits, awards limits, role targets.	Event-specific rating only after balancing.	P1
Live Arena	Synchronous room, spectators, timer, reactions, and bracket tools.	Yes in dedicated queues.	P2
Peak Auction	Shared player pool and budget bidding.	Dedicated event/rating.	P2

Table 16. Competitive mode portfolio and launch discipline.

Global ranking architecture

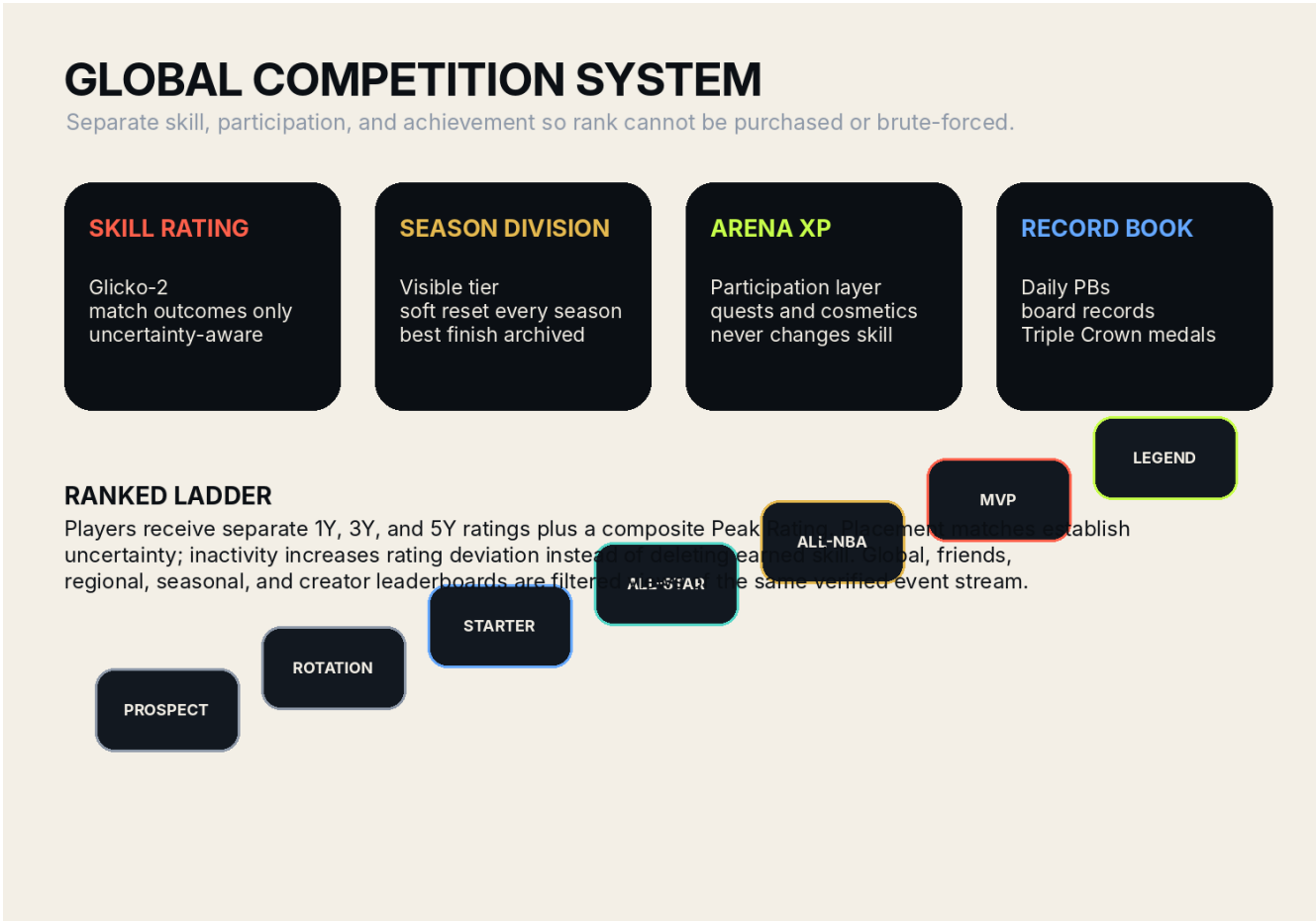


Figure 7. Four distinct layers: Glicko-2 skill, divisions, seasonal competition, and participation progression.

THE CORE SEPARATION

Skill rating answers “How strong are you against comparable opponents?” Seasonal points answer “What have you accomplished this season?” XP answers “How much have you engaged?” Record books answer “What exceptional thing did you do?” These must be separate.

Glicko-2 rating

Use Glicko-2 rather than a simple cumulative score. It represents rating, rating deviation (uncertainty), and volatility. New players are provisional; active, well-measured players move less dramatically; inactive players become less certain without being arbitrarily punished. Maintain independent ratings for 1Y, 3Y, and 5Y because the knowledge and strategy differ.

Rating field	Purpose	Product treatment
rating	Estimated skill in one queue.	Shown as division progress; exact number available in detailed profile.
rating deviation	Uncertainty around estimated skill.	Shown as “provisional” or “established,” not as a technical sigma on the main screen.
volatility	Expected consistency/change in performance.	Internal by default; useful for matchmaking and integrity review.
matches played	Evidence supporting the estimate.	Shown in queue profile and placement progress.

Rating field	Purpose	Product treatment
last rated activity	Controls uncertainty growth across rating periods.	Used for RD update; never sold as a punitive decay.

Table 17. How Glicko-2 fields translate into user experience.

Placements

- Require 7 completed, integrity-valid matches per queue before showing a division.
- During placements, match broadly but cap repeated opponents and repeated board templates.
- Show "Placement 3 of 7" and a performance summary, not a fake provisional rank.
- Unlock composite rank only after at least two queues are established; label it incomplete until all three are established.
- Returning players after a long break keep their historical rating but receive higher uncertainty and a short recalibration indicator.

Divisions

Division	Positioning	Presentation
Prospect	Entry and early placements settlement.	Simple crest; no shame language.
Rotation	Developing competence.	First seasonal cosmetic frame.
Starter	Above broad active-player median.	Visible profile badge and match intro.
All-Star	Strong global player.	Distinct board/result treatment and seasonal reward.
All-NBA	Elite, queue-specific mastery.	Numbered percentile and richer profile context.
MVP	Top competitive population band.	Public leaderboard eligibility and promotion animation.
Legend	Top fixed population or rating threshold with minimum activity.	Numbered global rank, verified seasonal record, audit scrutiny.

Table 18. Proposed division vocabulary. Final numeric thresholds should be set from live distribution, not invented before beta.

Match outcome design

A ranked duel compares two players on the same board. The primary outcome should be based on official lineup result, with deterministic tie-breakers. Do not award extra rating for playing faster unless a dedicated timed queue makes speed part of the advertised skill.

```
match_result = compare(official_score, tie_break_1, tie_break_2)
```

Suggested tie-breaks: board-relative draft efficiency, then fewer invalid/forced role placements; exact same outcome remains a draw.

- **Score normalization.** For mixed-board events, convert results to board percentiles or standardized values before aggregation.
- **No hidden handicaps.** Matchmaking can use rating and uncertainty, but the score formula is identical for every player.
- **Settlement.** A result is rated only after both submissions are complete, server-verified, and linked to the same immutable board snapshot.
- **Abort policy.** A player who abandons after seeing offers receives a loss after a clearly communicated grace rule; connectivity failures use limited protected aborts.
- **Rematches.** Allow one-click rematch on a new board. Cap repetitive pairings in rating windows to limit boosting.

Leaderboards

Board	Ranking key	Eligibility / reset
Global skill	Queue Glicko-2 rating, then RD and valid-match tie-breaks.	Minimum established status and recent rated activity; continuous.
Season	Season points from best bounded set of weekly results plus tournament placements.	Resets every 8-12 weeks; rating does not reset.

Board	Ranking key	Eligibility / reset
Daily board	Official score / percentile on one immutable board.	Closes after a fixed UTC settlement window; archive remains.
Friends	Same metric as selected global/season/daily view.	Private social graph; no separate scoring.
Regional	Global skill filtered by self-selected broad region.	Opt-in; never show precise location.
Creator event	Event-specific normalized total.	Explicit start/end and rules snapshot.
Record book	Highest score, best Triple Crown, longest valid win streak, rare achievements.	Versioned and separated by ruleset/model version where necessary.
Personal	Best score, percentile, rating, and streak by mode and rolling period.	Always available; designed to make progress attainable.

Table 19. Global and personal leaderboard system.

Seasonal structure

1. Use 8-12 week seasons. A season has a visual identity, one featured historical theme, a public schedule, and a finite championship path.
2. Rating persists between seasons. Seasonal points and cosmetic progression reset, with a soft placement checkpoint for awards but no forced grind back to skill level.
3. Count only a bounded number of strongest weekly performances toward seasonal points. This reduces raw-volume advantage while preserving reasons to play regularly.
4. Offer weekly qualifiers and one open championship event. The top ladder players receive seeding or byes, not exclusive access.
5. Archive every season with champions, model version, featured boards, distribution snapshots, and rule changes.

Triple Crown

Triple Crown should become PEAK3's signature championship. It consists of one 1Y Apex board, one 3Y Prime board, and one 5Y Foundation board. Each board is scored relative to its own population or simulation distribution; the normalized scores are combined so one unusually easy board does not dominate.

$$\text{triple_crown} = \text{mean}(z_{1Y}, z_{3Y}, z_{5Y}) - \text{consistency_penalty}$$

The consistency penalty, if used, must be small, published, and validated; a simple mean is acceptable for launch.

- Daily/weekly Crown: asynchronous, accessible to all, spoiler-safe sharing.
- Ranked Crown: pair users on the same three boards and settle only when all are complete.
- Creator Crown: public event link, branded landing page, audience leaderboard, and aggregate decision report.
- Season Finals: streamed or synchronous later, with locked clients and enhanced integrity controls.

Creative competitive modes

Mode	Distinctive mechanic	Why it has legs
The Gauntlet	Win consecutive boards across escalating constraints; one loss ends the run.	Creates dramatic runs and spectatable personal records.
Counterpick	Players alternate selecting one constraint that applies to both identical boards.	Adds strategic identity before the draft without changing fairness.
Peak Budget	Every exact window has a transparent salary; build five within a cap.	Makes undervalued peaks and model knowledge central.
Mirror Draft	Both see the same card triplets; once one player chooses a card, the opponent cannot take it.	High strategy and tension; ideal for synchronous P2.
Blind Resume Sprint	Identity-hidden comparisons contribute to a short streak and an accuracy leaderboard.	Fast, educational, and highly shareable.

Mode	Distinctive mechanic	Why it has legs
Era Relay	Five rounds each require a different era band, but roles remain flexible.	Broadens historical knowledge without pretending eras are directly simulated.
King of the Board	A champion lineup stays until another user beats it on the same board.	Persistent public target; easy creator/community ritual.
Decision Draft	Score includes final result plus prediction accuracy about which alternative would improve the roster.	Rewards understanding of the model, not memorized top cards.

Table 20. P1/P2 competitive concepts that remain native to PEAK3.

PART 07

Retention, progression, and social systems

Build repeatable reasons to return without manufacturing anxiety or turning the ladder into a grind contest.

Retention loop

trigger -> meaningful choice -> clear outcome -> progress -> social challenge -> next attainable goal

The daily board supplies the trigger. The draft supplies meaningful choice. The receipt supplies learning. Rating, personal records, and collections supply progress. Challenge links and rivals supply social energy. A visible next milestone supplies the return reason.

Progression layers

Layer	What it measures	Rewards	Must not do
Skill rating	Competitive performance against others.	Division, rank, ranked access, seasonal seeding.	Increase from payment, raw play count, or cosmetic XP.
Season points	Bounded accomplishments in a defined season.	Season track, qualifier seeding, archive status.	Reward infinite grinding; hide scoring rules.
XP / level	Participation, learning, and feature exploration.	Profile cosmetics, titles, noncompetitive customization.	Imply level equals skill.
Streak	Return ritual across valid local days.	Small cosmetic milestones, streak history, reserve token.	Remove rating, lock content, or shame users.
Achievements	Specific feats, knowledge, creativity, and rare outcomes.	Badges, trophy case, share cards.	Use purely repetitive "play 1,000 games" designs.
Collections	Discovery of players, seasons, eras, archetypes, and special board moments.	Card back designs, research shortcuts, optional goals.	Create pay-to-own scarcity around basic statistical access.
Personal records	Self-comparison independent of global standing.	Best score, percentile, streak, upset, role balance.	Erase records when a model changes; version them.

Table 21. Separate progression systems and their constraints.

XP design

XP should reward completion and exploration, not winning. A simple bounded schedule is enough: first daily completion, one ranked completion, one challenge completed, one formula layer explored, and one comparison saved. Repeated games can give sharply diminishing XP. This gives casual players progression while preserving the meaning of rating.

Action	Illustrative XP	Guardrail
First game of the day	100	Once per local day; server time and timezone safeguards.
First game in another window	40	Encourages 1Y/3Y/5Y discovery; once per day each.
Complete a received challenge	35	No XP for self-challenges or repeated same opponent farming.
Review full receipt / formula node	20	Award only once per new node category, then weekly cap.

Action	Illustrative XP	Guardrail
Ranked game after first	10 then diminishing	Daily soft cap; rating remains independent.
Publish a creator board with valid finishers	Event-based	Requires minimum unique participants and abuse checks.

Table 22. Illustrative XP schedule. Tune after observing behavior; never copy these values blindly into production.

Streak design

- **Definition.** Complete one qualifying Daily Peak or ranked match during the user's selected local day.
- **Reserve.** Earn one streak reserve after a healthy weekly cadence; cap storage at one or two.
- **Repair.** Allow a short, clearly bounded repair window through completion, not payment.
- **Timezone.** Store user-selected IANA timezone, limit frequent changes, and handle travel with an explicit safe change flow.
- **Presentation.** Celebrate 3, 7, 14, 30, 100, and 365; do not send escalating guilt language.
- **Competitive separation.** Streak has no direct effect on rating, board seed, matchmaking, or tools.

Achievement framework

Category	Examples
Knowledge	Correctly identify 10 Blind Resume winners; win with a non-obvious lower-score fit selection; complete an all-era relay.
Construction	Finish above the 95th percentile with no top-decile individual card; achieve elite coverage with five different archetypes.
Competition	Win a rated match as a rating underdog; achieve a clean Triple Crown; earn promotion in all three queues.
Discovery	Inspect every formula branch; compare peaks from five decades; download and cite a data release.
Community	Beat a creator board; host a challenge with 100 valid finishers; contribute an accepted data issue or methodology discussion.
Rare / seasonal	Finish #1 on a daily board; win a season tournament; hold King of the Board for 24 hours.

Table 23. Achievement categories focused on meaningful behavior rather than empty volume.

Social graph and competition

- **Follow, not forced friendship.** Let players follow creators and peers; mutual follows can unlock friends-only comparisons.
- **Rivals.** Suggest opt-in rivals near the player's rating and activity pattern. Show recent head-to-head record and invite rematches.
- **Challenge inbox.** Organize pending, completed, expired, and creator challenges. Notifications must be configurable.
- **Activity feed.** Show meaningful events only: promotions, records, tournament results, and direct challenge outcomes.
- **No open chat at launch.** Use structured reactions and share text. Open text substantially increases moderation and safety scope.
- **Privacy.** Profile, region, activity, match history, and follows each need public/followers/private controls.

Notifications

Trigger	Channel default	Frequency rule
Direct challenge received	In-app; optional push/email	Immediate, but bundle duplicates.
Ranked match settled	In-app; optional push	Immediate when user is not active.
Daily board available	Opt-in push/email	One per day maximum, local preferred time.
Streak at risk	Opt-in only	One reminder, never late night.

Trigger	Channel default	Frequency rule
Promotion / record beaten	In-app; optional push	Immediate for promotion; batched for noncritical record changes.
Season event	In-app and optional email	Schedule notice, start reminder, final-hours reminder with controls.

Table 24. Notification rules that support return without becoming noise.

PART 08

Scoring, lineup logic, and fairness

PEAK3 evaluates individual peaks. A separate lineup layer must be bounded, testable, and honest about what it does not know.

Three score layers

Layer	Purpose	Authority status
PEAK3 individual/window score	Quantifies one exact season or contiguous peak window using the established model.	Canonical PEAK3 model; versioned and validated through the project's methodology/tests.
Lineup coverage and synergy	Evaluates whether five selected peak profiles form a coherent hypothetical roster.	New experimental model; must be labeled, validated, and separately versioned.
Board-relative performance	Measures how well a user drafted given the exact available board.	Game-system metric; computed from simulation or population outcomes.

Table 25. Do not blend the authority status of individual and lineup models.

Prototype lineup equation

official_lineup_score = 0.78 * talent + 0.14 * coverage + 0.08 * synergy

Research hypothesis for simulation and testing only. Final weights require validation and public documentation.

Talent should dominate because PEAK3's intellectual value is the foundation. Coverage makes severe roster holes matter. Synergy recognizes bounded complements and conflicts. No fit model should transform a mediocre collection into an all-time team, and no subjective archetype label should erase overwhelming individual quality.

Talent aggregation

A simple arithmetic mean permits five redundant superstars to add perfectly. A strict weakest-link function overpunishes role players. Begin with an ordered, diminishing contribution model: the strongest peak contributes most, subsequent cards contribute slightly less, and every role still matters. Normalize by window and model version before aggregation.

talent = sum(w_i * normalized_peak_i), where w_1 >= ... >= w_5 and sum(w_i)=1

Order by contribution after role assignment, not by draft round. Test against alternative means and sensitivity analyses.

Coverage

Represent each card as a vector over the eight Lineup DNA dimensions. Aggregate the roster with saturation so duplicate strength has diminishing returns. Compute a coverage score that rewards adequate minimums and broad quality. A harmonic or geometric component can penalize catastrophic holes without turning every lineup into a forced template.

coverage = blend(mean(saturated_dimensions), harmonic_floor(dimensions))

All transformations, floors, and caps must be visible in the Lab.

Synergy

Synergy should be a small signed adjustment based on explicitly defined interactions. Positive examples include elite creator plus scalable off-ball shooting, rim pressure plus spacing, and perimeter containment plus interior protection. Negative examples include extreme usage redundancy, multiple non-spacing frontcourt roles, no primary creator, and no viable anchor. Keep the total effect bounded, such as within +/- 5 to 7 displayed points.

Peak Receipt

Receipt item	Example	Calculation requirement
Talent core	"Your five peaks ranked in the 99.1st percentile for raw individual quality."	Direct from normalized, versioned peak inputs.
Coverage gain	"Adding 2014 Kawhi raised perimeter defense and off-ball scalability."	Counterfactual before/after on the same roster state.
Conflict penalty	"Two high-control creators created a -1.8 role-overlap adjustment."	Named rule, cap, inputs, and model version.
Weakness	"Interior defense remained below the 31st percentile of completed lineups."	Compare with board or global reference population.
Decision opportunity	"Round 4 alternative X would have improved anchor coverage by 2.1."	Server-replay of valid unselected choice; label as model estimate.
Uncertainty / completeness	"One historical card has limited defensive context."	Surface data-quality flags; avoid pretending precision is equal across eras.

Table 26. Every score explanation should map to a reproducible computation.

Board-relative Draft Efficiency

Global score measures the final roster. Draft Efficiency measures how effectively the user navigated a particular board. It is useful because some boards are richer or harder than others. It must not be approximated with a naive "best card every round" strategy; the optimum is a whole-draft search with role, tool, and path constraints.

$$\text{draft_efficiency} = (\text{user_score} - \text{board_floor}) / (\text{board_optimum} - \text{board_floor})$$

Clamp only for presentation; retain raw values. Use exact search when feasible or validated beam/Monte Carlo search with confidence metadata.

- **Board floor.** Use a defensible lower reference such as the 5th percentile of valid simulated policies, not the mathematically worst absurd lineup.
- **Board optimum.** Search the full offer tree including Hold/Reframe rules. Record the solver and version.
- **Ranked use.** Final official lineup score should decide direct matches; Draft Efficiency is a tie-breaker and analytical measure until validated.
- **Public language.** Call it Board Efficiency or Draft Efficiency and explain it as "how much of this board's available value you captured."

Fairness and historical context

- ☐ Document role eligibility separately from score inputs and provide a correction workflow.
- ☐ Surface missing-data and era-coverage flags directly on cards and receipts.
- ☐ Do not infer modern three-point shooting, switching, or matchup portability beyond the evidence without labeling the assumption.
- ☐ Maintain separate distributions for 1Y, 3Y, and 5Y and never compare raw scores across incompatible model versions without calibration.
- ☐ Test whether specific eras, positions, teams, or archetypes systematically receive easier role completion or more favorable boards.
- ☐ Run blind human reviews with knowledgeable evaluators and record disagreement, not merely agreement with the model.
- ☐ Publish model changes, affected records, and backward-compatibility treatment before deploying them to ranked play.

Validation program

Test family	Question	Minimum evidence before ranked launch
Synthetic invariants	Does improving one input behave monotonically where expected?	Automated property tests for caps, missingness, ordering, and role rules.
Historical face validity	Do outcomes match defensible basketball intuition without being forced to consensus?	Curated set of hundreds of lineups and counterfactuals reviewed by model authors.
Expert pairwise review	Can knowledgeable reviewers choose between lineup pairs, and how often does the model agree?	Blind panel, inter-rater agreement, documented disagreements.
Distribution audit	Do roles/eras/positions receive systematic score advantages?	Slice reports and launch thresholds.
Adversarial search	Can an optimizer exploit a tag, cap, or interaction to produce absurd top lineups?	Automated optimizer plus manual red-team review.
Live calibration	Do high-rated players consistently outperform lower-rated players on unseen boards?	Beta rating calibration and predictive accuracy report.

Table 27. Validation program for the new lineup layer and competitive system.

PART 09

The interactive formula explorer

Turn methodology into a navigable visual object: zoom from the whole model to components, transformations, exact inputs, and source records.

Product goal

The formula explorer should feel like exploring a map, not reading a wall of equations. A casual fan can understand the five major ingredients. An analyst can descend to every transformation, normalization, cap, source, and missing-data rule. A player can begin from a specific card or game receipt and trace the exact path that produced the number.

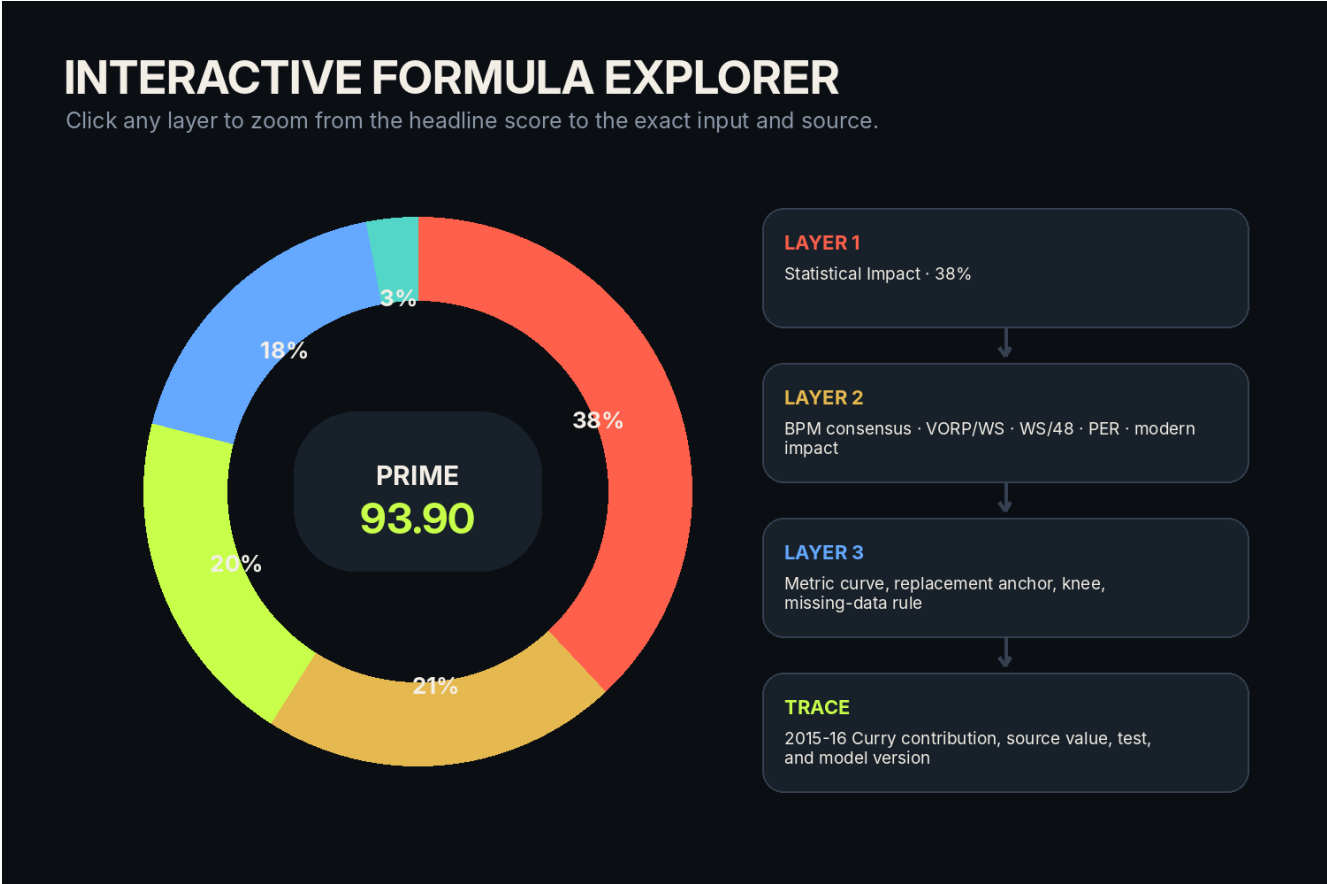


Figure 8. Formula Explorer structure: overview rings, component nodes, sub-metrics, transformations, source records, and a selected-player trace.

Four zoom levels

Level	Default visual	What the user learns	Primary interaction
1. Model overview	Five weighted arcs or connected regions around final PEAK3 score.	The broad philosophy and official component weights.	Select a component; toggle performance-only vs full score.

Level	Default visual	What the user learns	Primary interaction
2. Component anatomy	Node tree or flow diagram for one component.	Which sub-metrics contribute, how missing metrics are handled, and where caps apply.	Expand nodes; compare components; inspect definitions.
3. Metric transformation	Interactive curve with distribution and selected player marker.	How raw data becomes era-relative standardized value and then component contribution.	Drag raw value, switch season, show percentile/z-score/cap.
4. Provenance trace	Exact player-season input rows, source citation, processing step, version, and quality flag.	The evidence chain behind the displayed result.	Open source record, download row, copy citation, report issue.

Table 28. Progressive levels in the interactive methodology experience.

Interaction model

- **Zoom and focus.** Scroll/pinch or click a component to smoothly focus it; preserve breadcrumb and back path. Do not require precision panning on mobile.
- **Cross-highlighting.** Hovering or focusing a component highlights its contribution in the equation, selected card, career chart, and comparison table.
- **Player trace.** Search or enter from a card to animate the exact selected peak through each layer.
- **Compare trace.** Overlay two peaks, using color plus shape/line style so differences remain accessible.
- **What-if sandbox.** Allow users to adjust weights or disable a layer, but label the output “custom, unofficial” and never confuse it with ranked scoring.
- **Narrated mode.** Offer a short step-through explanation for first-time visitors and educators.
- **Deep links.** Every node, player trace, metric curve, and model version receives a shareable URL.
- **Reduced motion.** Replace zoom transitions with instant focus changes while preserving hierarchy and context.

Formula Explorer screen anatomy



Figure 9. Formula Explorer direction: central visual canvas, persistent breadcrumb, selected peak trace, explanatory side panel, and data/provenance controls.

Region	Required behavior
Top bar	Model version selector, performance-only/full toggle, selected player/window, share, and methodology status.
Canvas	Responsive SVG/canvas hierarchy with keyboard navigable nodes and semantic fallback list.
Context panel	Plain-language definition, official equation, transformations, assumptions, limits, and links to tests/data.
Trace rail	Raw value → normalized value → transformed value → weight → contribution; compare mode supports two traces.
Footer drawer	Source rows, provenance, data completeness, API example, download, and issue report.

Table 29. Formula Explorer layout contract.

Visual encoding

Meaning	Encoding
Statistical Impact	Lime family; sharp angular path.
Two-Way Production	Blue family; paired shape or balanced axis.
Recognition	Gold family; badge/medallion motif used sparingly.
Postseason	Coral family; bracket/path motif.
Team Success	Teal family; small bounded segment.
Positive contribution	Filled or rising visual with explicit + label.

Meaning	Encoding
Negative/cap/missingness	Pattern, boundary, and text icon; never red/green alone.
Unofficial custom output	Dashed border, neutral violet/gray treatment, persistent "Custom" label.

Table 30. Semantic visual encoding for the model.

Implementation approach

- **Schema-driven graph.** Generate the explorer from a versioned methodology JSON graph rather than hard-coding every screen. Nodes contain IDs, labels, equations, dependencies, transforms, source fields, explanation copy, and links.
- **D3 hierarchy/zoom.** Use D3 for custom hierarchy, transitions, scales, and zoom behavior. Keep rendering controlled by React and maintain semantic DOM fallbacks.
- **Observable Plot for distributions.** Use Plot for histograms, scatterplots, percentile bands, and player markers where conventional statistical graphics are sufficient.
- **Static first render.** Server-render the overview, definitions, and accessible outline. Hydrate interaction progressively.
- **Version bundle.** The page loads one immutable methodology bundle matching the selected model version; do not mix live metadata with an old score.
- **Screenshot/export.** Provide SVG/PNG export with title, selected peaks, model version, PEAK3 attribution, and canonical URL.

Content requirements per node

- ☐ Stable node ID and model version.
- ☐ One-sentence plain-language purpose.
- ☐ Formal equation or transformation.
- ☐ Inputs, units, expected range, missing-data behavior, and caps.
- ☐ Why the design choice exists and what alternative was rejected.
- ☐ Known limitations and historical coverage.
- ☐ One selected-player example and one counterexample.
- ☐ Source/provenance link and downloadable fields.
- ☐ Tests/invariants that protect the node.
- ☐ Change history and affected versions.

PART 10

PEAK3 Index, comparisons, and public data

The game creates attention; the Index and data platform convert that attention into lasting authority and utility.

Index home

The Index home should feel like an editorial statistical reference rather than a spreadsheet dump. Lead with the three canonical windows, a search box, one compelling visual story, and direct access to full tables. Every headline value should open into an exact season/window page and a formula trace.

Leaderboard requirements

Control	Behavior
Window	1Y, 2Y, 3Y, 5Y, 10Y, and custom contiguous window where supported.
Score type	Full PEAK3; performance-only; regular-season performance; component views.
Era	Season range or decade; disclose dataset start and coverage.
Role / position	Use canonical historical positions plus functional role filters.
Team	Filter exact selected seasons/windows by franchise representation.
Postseason / awards	Filter by playoff stage, championship, MVP/DPOY/All-NBA status without conflating with score.
Data completeness	Option to include/exclude windows with limited modern metrics or defensive context.
Model version	Default current; historic versions available when records differ.
Export / share	Download current filtered rows, copy canonical link, generate labeled image.

Table 31. Required filters and controls for canonical leaderboards.

Player page

Module	Required contents
Header	Name, era, primary roles, career span, best 1Y/3Y/5Y, current model version, compare/play actions.
Peak Mountain	Every season as a point/line; highlighted best 1Y, 3Y ridge, and 5Y plateau; selectable window.
Window cards	Best windows by duration with score, rank, seasons, components, and uncertainty/completeness flags.
Season table	Raw and derived fields, component scores, postseason context, awards, team, rank, links.
Score anatomy	Selected window's formula fingerprint and contribution trace.
Peer set	Similar profiles, adjacent rankings, same-era/role comparisons, and "often compared" links.
Game history	How often the peak appears in official boards; selection rate; win contribution; not used to alter canonical score.
Citations	Model version, data release, methodology URL, stable player/window IDs, citation copy.

Table 32. Player page modules.

Peak Mountain signature visualization

Peak Mountain should become the recognizable visual symbol of the Index. Individual seasons form the career terrain. The best one-year peak is a summit marker. The best three-year window becomes a highlighted ridge. The best five-

year window becomes a broader plateau. Tooltips show scores and components; selecting a region updates the page. Exported visuals include model version and exact window definition.

- Use one consistent vertical score scale per score type, with a clear toggle between absolute display score and career-relative view.
- Do not smooth the underlying season points so strongly that actual values become visually false.
- Show missed seasons and unavailable data as gaps or patterned markers, not zero values.
- Provide table and text alternatives with the same selected-window state.
- Allow comparison overlays, but limit default overlays to two or three players to preserve legibility.

Compare

Compare should support up to four exact seasons/windows and answer “why does one rank higher?” rather than presenting parallel cards with no conclusion.

1. Summary strip: score, rank, window, performance-only/full difference, and data-quality status.
2. Component slope chart or difference bars with the selected comparison baseline.
3. Statistical profile: era-relative values, raw values, workload, role, and playoff context.
4. Contribution explanation: the largest three factors separating each pair.
5. Counterfactual toggles: remove recognition, remove team success, regular season only, or custom unofficial weights.
6. Share and export: landscape analysis card, vertical social card, CSV rows, canonical URL, citation.

Season and era pages

- **Season pages.** Top peaks in that season; distribution; league environment; awards; postseason; normalization inputs; notable model disagreements.
- **Era pages.** Decade/era leaderboards, score distributions, data coverage, role trends, and representative peaks without asserting a simplistic “best era.”
- **Franchise pages.** Best exact seasons/windows produced while representing the team; historical cards; Arena franchise boards. Use text identity unless trademark strategy permits more.
- **Record pages.** Highest component values, largest regular/playoff gaps, strongest non-award profiles, best lower-seeded postseason runs, and other editorially defensible records.

PEAK3 Lab data explorer

The browser-based Lab should let analysts filter, group, chart, and export without writing code. Use precomputed Parquet/Arrow or API-backed subsets depending on file size. Common queries should be instant and shareable.

Capability	P0/P1 behavior
Table builder	Select fields, filter, sort, group, paginate, save query, download result.
Chart builder	Scatter, line, bar, histogram, rank, and distribution views; highlight players/windows.
Calculated view	Choose existing official derived fields only in P0; custom calculations in P1 sandbox.
Model comparison	Compare ranks and scores across model releases; show movers and reason categories.
Query URL	Serialize filters, fields, chart encodings, and version in a short shareable state.
Export	CSV, Parquet, JSON, SVG/PNG with definitions and version metadata.
Notebook examples	Python, R, JavaScript, DuckDB SQL, and curl examples tied to a release.

Table 33. Public data explorer capabilities.

Public release structure

```
release/{semver}/ = immutable data + schema + provenance + methodology + changelog
                    + checksums
```

Artifact	Purpose
player_seasons.parquet / .csv	One row per player-season with canonical IDs, raw inputs, component outputs, completeness, and provenance keys.
peak_windows.parquet / .csv	One row per evaluated contiguous window with duration, seasons, weighting, score outputs, rank, and version.
players.json	Canonical player metadata, aliases, career span, role eligibility, and stable ID.
leaderboards/*.csv	Convenience snapshots for common 1Y/3Y/5Y/full/performance-only views.
methodology.json	Machine-readable formula graph used by Formula Explorer.
data_dictionary.json	Field definition, type, units, allowed values, missingness meaning, and authority status.
provenance.csv	Source and transformation lineage by field or row group, respecting redistribution constraints.
checksums.txt	Integrity hashes for reproducibility.
CHANGELOG.md	Added data, corrections, methodology changes, rank impact, and migration notes.

Table 34. Versioned public data release contract.

API surface

Endpoint	Purpose	Caching / auth
GET /api/v1/leaderboards	Filtered canonical leaderboard with cursor pagination.	Public; CDN-cache by query and model version.
GET /api/v1/players/{id}	Player metadata, best windows, links.	Public; long cache for immutable versions.
GET /api/v1/windows/{id}	Exact window, component breakdown, source flags.	Public; immutable/versioned cache.
GET /api/v1/compare	Server-generated comparison summary for selected windows.	Public; bounded inputs and cached.
GET /api/v1/releases	Release metadata and current version.	Public.
GET /api/v1/boards/{public_id}	Public board metadata and settled aggregate results.	Public after spoiler window; offers require game authorization.
POST /api/v1/games	Create an anonymous or authenticated game session.	Rate-limited; CSRF/origin checks.
POST /api/v1/games/{id}/actions	Submit authoritative draft action with state token.	Authenticated session; idempotency required.
GET /api/v1/me/*	Profile, ratings, progress, matches, follows, settings.	Authenticated; RLS.

Table 35. Initial API boundaries.

PART 11

Visual design, motion, accessibility, and performance

The visual system should feel like premium sports editorial, archival basketball, and a modern competitive interface - not a generic neon dashboard.

Design concept: Arena Archive

Combine the authority of an archival scouting file, the immediacy of an arena scoreboard, the restraint of a premium sports publication, and the precision of a research tool. Exact peak cards are the hero object. Large numerals, structured grids, thin technical lines, and selective high-energy accents create identity without turning every screen into a video game lobby.

Color system

Token	Hex	Role
Ink	#0B0F14	Primary dark background and high-contrast text.
Panel	#121820	Elevated surfaces, cards, navigation, result modules.
Paper	#F3EFE6	Warm light text and editorial background alternative.
Peak Lime	#C7FF4A	Primary action, selected state, score highlight, model energy.
Apex Coral	#FF604B	1Y identity, warnings, competitive intensity.
Prime Gold	#E5B94F	3Y identity, medals, recognition component.
Foundation Blue	#65A8FF	5Y identity, analytical depth, information state.
Context Teal	#53D6C7	Team/context layer, positive secondary state.

Table 36. Proposed core palette. Validate contrast in final component combinations.

Typography

- **Display.** Use a condensed or tightly spaced contemporary display face for major ranks, scores, titles, and game moments. It should feel editorial rather than futuristic novelty.
- **Interface/body.** Use a highly legible sans serif such as Inter or a licensed equivalent, with tabular numerals for scores and tables.
- **Data.** Use tabular numbers and a restrained monospace face only for formulas, IDs, and code examples.
- **Hierarchy.** Large number + concise label is the primary motif. Avoid all-caps paragraphs and overly small gray metadata.
- **Licensing.** Use open or properly licensed fonts; self-host subsets to reduce layout shift and external dependencies.

Grid, shape, and card language

Element	Rule
Layout grid	4 px base rhythm; 12-column desktop, 6 tablet, 4 phone; maximum editorial text measure around 65-75 characters.
Corners	Moderate 12-18 px radii for cards; smaller 8-10 px for controls; avoid pill shapes for every element.
Borders	Thin low-contrast technical lines; strong accent only for selected, promoted, or critical state.
Depth	Prefer borders, inset layers, and restrained shadows; avoid floating glass panels everywhere.
Peak cards	Fixed information hierarchy, distinct window accent, prominent exact seasons, role tags, score fingerprint.
Charts	Neutral grid and labels; one active highlight; team/era colors only when semantically required.

Element	Rule
Textures	Extremely subtle grain or printed-paper treatment; never reduce text contrast or performance.

Table 37. Visual-system rules.

Motion system

Motion	Purpose	Timing guidance
Card deal	Establish new round and spatial relationship.	180-260 ms, stagger ≤ 60 ms, transform/opacity only.
Selection lock	Confirm authoritative placement and move card to lineup.	160-220 ms; haptic where supported.
DNA update	Show which roster dimensions changed.	250-400 ms with previous-value anchor; no endless pulsing.
Score assembly	Explain talent, coverage, synergy, and final result.	Sequential 600-1000 ms total; skip/instant in reduced motion.
Promotion	Reward a rare status transition.	1.2-1.8 s maximum; never on every win.
Formula zoom	Maintain hierarchy as a node becomes focus.	220-400 ms; instant alternative.
Skeleton/loading	Communicate real progress or stable layout.	No artificial delay; avoid shimmering large areas.

Table 38. Purposeful motion timing and constraints.

Performance budget

Metric / budget	Target
Largest Contentful Paint	≤ 2.5 seconds at the 75th percentile on target mobile traffic.
Interaction to Next Paint	≤ 200 ms at the 75th percentile; draft actions should feel immediate locally while awaiting server authority.
Cumulative Layout Shift	≤ 0.1 ; reserve card, font, chart, and ad-free layout space.
Initial JS on game route	Set an explicit budget; aim under roughly 180-220 KB compressed for critical interactive code, then measure.
Images	Original vector/optimized raster assets; responsive sizes; no player-photo dependency.
Animation	Transform and opacity on compositor; avoid layout-triggering properties in active game transitions.
Third-party scripts	None on critical path; analytics consent and lazy initialization.
Offline resilience	Cache static data and active board shell; safely recover from transient network loss without accepting invalid actions.

Table 39. Launch performance contract. Budgets should be enforced in CI and real-user monitoring.

Accessibility launch checklist

- ☐ Every game action is reachable and operable by keyboard with a clear, persistent focus indicator.
- ☐ Cards announce player/window, role eligibility, selected state, tool state, and relevant score band in a logical order.
- ☐ Drag-and-drop has button/menu alternatives; never require pointer precision.

- ❑ Color is never the sole encoding for window, score direction, component, selected state, or missing data.
- ❑ Text and non-text contrast are tested in actual tokens and component states.
- ❑ Reduced-motion preference replaces nonessential transitions and retains all information.
- ❑ Charts have summaries, data tables, selected-point descriptions, and keyboard exploration where interactive.
- ❑ Live regions announce authoritative action confirmation, round transition, score completion, and errors without chatter.
- ❑ Touch targets, zoom, orientation, responsive reflow, and text scaling meet WCAG 2.2 expectations.
- ❑ Automated axe checks are supplemented with keyboard, screen-reader, low-vision, and motion-sensitivity manual testing.

Design-reference workflow

Do not copy visual treatments from one fashionable site. Build a reference board by behavior: editorial hierarchy, game-state clarity, statistical storytelling, chart interaction, motion, responsive card systems, and accessible focus. Useful starting resources include Apple Human Interface Guidelines for motion and hierarchy, Material Design for state/feedback patterns, The Pudding for narrative data experiences, D3 and Observable Plot for visualization, and web.dev for performance budgets. The resource appendix links each reference to a specific implementation question.

PART 12

Technical architecture and data pipeline

Keep the Python model authoritative, generate immutable artifacts, and make the web game fast, replayable, and independently scalable.

Recommended stack

Layer	Recommendation	Reason
Web application	Next.js App Router + TypeScript; server components for data-heavy pages; client islands for game/chart interaction.	Strong routing, SSR/streaming, metadata, caching, and deployment ecosystem.
Styling / components	Tailwind or token-driven CSS plus custom PEAK3 component library; Storybook optional but valuable.	Rapid implementation with centralized tokens and visual regression.
Visualization	D3 for custom hierarchy/interaction; Observable Plot for conventional statistical charts.	Precise custom formula experience without hand-building every chart primitive.
Auth / database	Supabase Auth + PostgreSQL + Row Level Security.	Fast account launch, relational integrity, policies, and manageable operational scope.
Realtime	Supabase Realtime initially for match settlement/notifications; dedicated channel service only if scale demands.	Avoid premature distributed systems while preserving live updates.
Cache / queues	Managed Redis-compatible cache and durable job queue.	Leaderboards, rate limits, board locks, settlement jobs, and deduplication.
Model engine	Existing Python PEAK3 pipeline remains canonical; export versioned artifacts.	Prevents JavaScript reimplementations drift and supports reproducibility.
Object storage	S3-compatible storage for immutable releases, images, and exports.	CDN distribution, checksums, lifecycle, and public download URLs.
Observability	OpenTelemetry traces/metrics, structured logs, product analytics, Web Vitals, error tracking.	One request/game trace across web, API, jobs, and model version.
Testing	Pytest; Vitest/Jest; Playwright E2E, visual, and accessibility; property-based scoring tests.	Protect formulas, game state, rendering, and critical journeys.

Table 40. Recommended implementation stack.

System architecture

REFERENCE ARCHITECTURE

CLIENT

Next.js web app · server components · responsive game UI · D3/Plot

APPLICATION

Public data routes · game service · auth · profile · share renderer

COMPETITION

Board generator · scoring verifier · matchmaking · Glicko-2 · anti-cheat

DATA

Supabase Postgres/Auth/Realtime · Redis for queues/rate limits · object storage

MODEL

Existing Python PEAK3 engine · deterministic exports · versioned Parquet/JSON

OBSERVE

OpenTelemetry · errors · web vitals

DELIVER

Vercel/CDN · CI/CD · signed data releases

Figure 10. Reference architecture: versioned PEAK3 model artifacts feed public Index/Lab reads and server-authoritative Arena game services.

Repository structure

monorepo = engine + generated artifacts + web app + shared schemas + lineup model + infrastructure

Path	Responsibility
engine/	Existing PEAK3 Python scoring, data ingestion, validation, window computation, and command-line reports.
engine/lineup_research/	Experimental card profiles, Lineup DNA, coverage/synergy modeling, solver, and validation notebooks/tests.
generated/{version}/	Immutable web-ready JSON/Parquet, formula graph, public release files, board pool metadata, and checksums.
apps/web/	Next.js application, route handlers/BFF, game client, Index, Lab, profile, admin surfaces.
packages/ui/	Design tokens, accessible primitives, cards, charts, feedback, navigation, and documentation.
packages/schemas/	Generated TypeScript/Python contracts, Zod/JSON Schema/OpenAPI definitions, IDs, enums.
packages/game-core/	Deterministic board/rules logic shared where safe; client prediction only, server remains authoritative.
packages/telemetry/	Event names, trace helpers, privacy filters, correlation IDs, and observability conventions.

Path	Responsibility
infra/	Database migrations, RLS policies/tests, deployment config, queues, storage, monitoring, and runbooks.
docs/	Methodology, architecture decisions, game rules, model cards, security, data dictionary, and changelog.

Table 41. Recommended monorepo boundaries.

Model-to-web pipeline

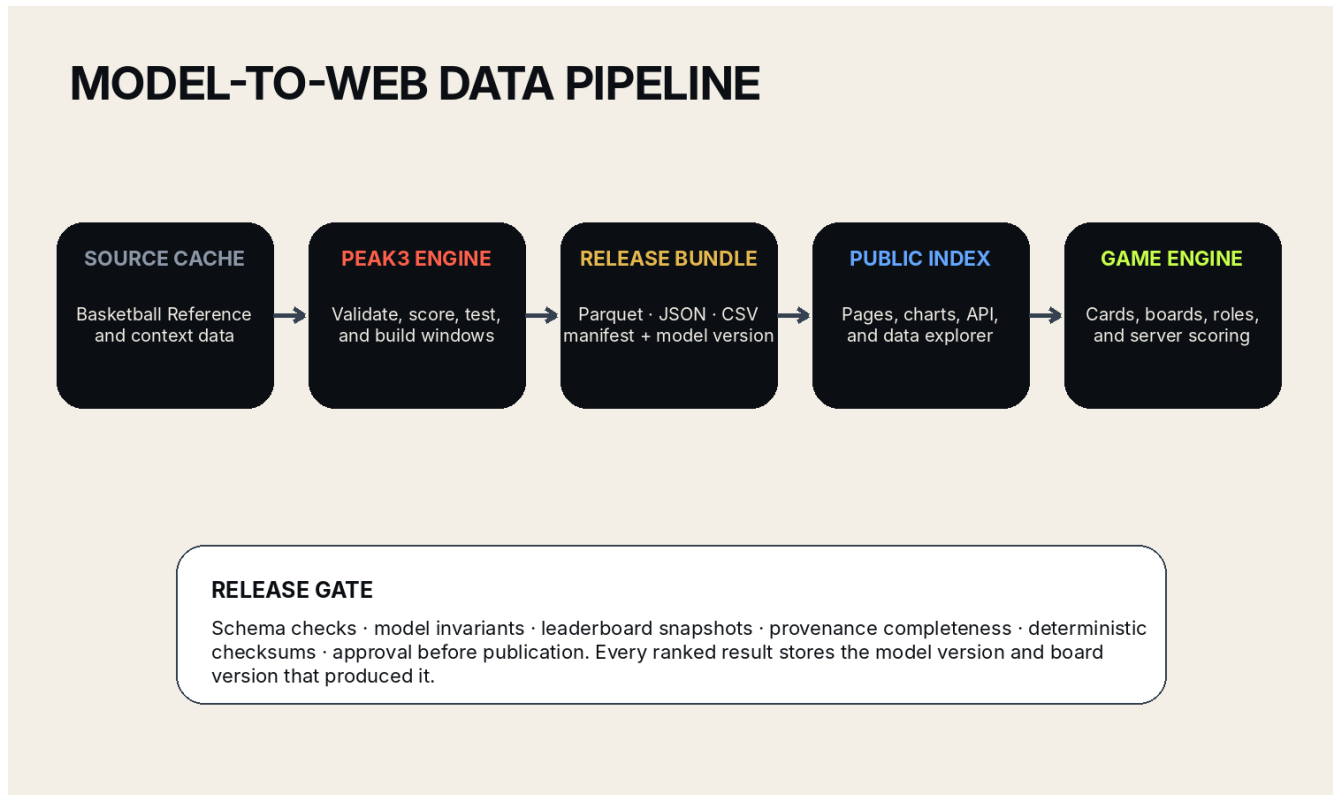


Figure 11. Data pipeline: source ingestion → PEAK3 validation → immutable release → web projections → board generation → game/result snapshots.

1. Acquire and cache source data under documented terms and provenance rules.
2. Validate schemas, player/team/season identity, duplicate handling, and expected coverage.
3. Run canonical PEAK3 engine and all formula/property/regression tests.
4. Generate player-season rows, window rows, leaderboards, model graph, definitions, and completeness metadata.
5. Generate stable IDs and release manifest; compute checksums and impact diff against prior version.
6. Publish release candidate to staging object storage and seed staging database projections.
7. Run visual/data smoke tests, rank-diff review, formula trace verification, and board simulations.
8. Approve immutable semantic version; publish files/API metadata; deploy web compatibility flag.
9. Ranked queues opt into the new version only at a scheduled season/rules boundary unless correcting a critical error.
10. Retain old artifacts indefinitely for game replay, citations, and historical comparisons.

Read-path design

- **Index pages.** Prefer static generation or cached server rendering from versioned artifacts/database projections. Cache keys always include model version and filters.
- **Formula.** Load small overview graph first; lazy-load selected node traces and source records.
- **Game.** Create server session, send only current legal offers/state, optimistically animate locally, then reconcile authoritative state.
- **Leaderboards.** Precompute common global/daily/season views; use cursor pagination and periodic/streaming updates instead of expensive live rank queries.
- **Profile.** Read through RLS-protected PostgreSQL views; aggregate history asynchronously.

- **Public API.** Separate immutable versioned endpoints from current aliases; enforce quotas and cache public GETs.

Deployment environments

Environment	Purpose	Data / access rule
Local	Fast engine, UI, and state-machine development.	Synthetic or sanitized fixtures; local Supabase/Postgres; deterministic seeds.
Preview	Per-branch visual/product review.	Isolated test project or schema; no production secrets/user data.
Staging	Release-candidate model, migrations, load tests, tournaments rehearsal.	Production-like data projections; allowlisted accounts; separate auth keys.
Production	Public Index/Lab/Arena.	Immutable releases; audited migrations; feature flags; monitored rollback.

Table 42. Environment separation and data rules.

PART 13

Database, API, realtime, and model-version contracts

The schema must preserve every official game as a reproducible event, not merely store a mutable final score.

Core database domains

Domain	Key tables / entities
Identity	profiles, handles, profile_settings, follows, blocks, notification_preferences, sessions/devices.
Model registry	model_versions, data_releases, ruleset_versions, card_pool_versions, methodology_nodes.
Basketball index	players, teams, seasons, player_season_projection, peak_windows_projection, ranks_projection, provenance_refs.
Games	boards, board_offer_nodes, games, game_actions, lineups, lineup_cards, game_results, result_receipts.
Competition	queues, ranked_matches, ratings, rating_events, divisions, seasons, season_points, tournaments, entries.
Progression	xp_events, levels, streaks, streak_reserves, achievements, achievement_unlocks, collections, personal_records.
Social / creator	challenges, creator_events, event_boards, event_entries, reactions, reports, moderation_actions.
Operations	idempotency_keys, integrity_flags, audit_log, feature_flags, notification_outbox, jobs, webhook_events.

Table 43. Core relational domains.

Critical entity contracts

Entity	Immutable fields	Mutable fields
model_version	semantic version, formula graph hash, code commit, input release, published timestamp.	status/deprecation note only.
ruleset_version	mode, round/tool rules, scoring layer versions, state-machine hash.	status only.
board	seed, mode, pool version, model/ruleset versions, offer-tree hash, publication checks.	visibility/settlement status.
game	board ID, owner/session identity, initial state hash, queue type, creation time.	current state, completion/abandon status.
game_action	sequence, action type/payload, previous hash, resulting hash, server timestamp.	Never mutate; append corrections as audit events.
game_result	final lineup snapshot, official score, receipt version, board metrics, integrity status.	Settlement/rating linkage only.
rating_event	before/after rating, RD, volatility, opponent/result, algorithm version, reason.	Never mutate; compensating event for correction.

Table 44. Immutable fields required for replay and trust.

Representative schema

Table / selected columns	Notes
model_versions(id, semver, engine_commit, input_release_id, formula_hash, status, published_at)	Referenced by every window, board, receipt, and public citation.
ruleset_versions(id, name, semver, state_machine_hash, lineup_model_version, config_json)	No ranked game begins without an explicit ruleset.
boards(id, public_id, seed, mode, model_version_id, ruleset_version_id, pool_version_id, offer_tree_hash, published_at)	Seed is not necessarily public before spoiler/settlement.
games(id, board_id, user_id nullable, anon_session_id nullable, queue, status, state_hash, started_at, completed_at)	Anonymous sessions can later be claimed with secure merge rules.
game_actions(id, game_id, seq, action_type, payload_json, prev_hash, next_hash, idempotency_key, server_at)	Unique(game_id, seq); append only.
lineup_cards(result_id, slot, window_id, draft_round, card_snapshot_json)	Snapshot protects old results when display metadata changes.
game_results(game_id, score, talent, coverage, synergy, percentile, efficiency, receipt_json, integrity_status)	Numeric precision and rounding rules documented.
ratings(user_id, queue, rating, rd, volatility, matches, algorithm_version, updated_at)	Unique user+queue.
rating_events(id, user_id, match_id, queue, before_json, after_json, created_at)	Ledger for audit/rebuild.
leaderboard_snapshots(id, board/season/queue scope, generated_at, model/ruleset versions, object_url)	Fast historical pages and audit.

Table 45. Representative schema fields. Full DDL belongs in migrations and schema docs.

Row Level Security

- **Profiles.** Public fields available through a safe view; private settings, email/auth metadata, and region details restricted to owner/admin roles.
- **Games.** Owner can read private in-progress game; public can read only permitted completed snapshots; opponents receive only information allowed by match state.
- **Actions.** Clients never directly insert authoritative actions into base tables. Use server route/function with validation and service permissions.
- **Ratings.** Public leaderboard view exposes necessary rank fields; rating event ledger is owner/admin or summarized.
- **Follows/blocks.** Owner-managed; blocked users cannot create direct challenges or follow interactions.
- **Admin.** Use explicit application roles and audited server paths; never rely on hidden UI as authorization.
- **Testing.** Every policy receives allow/deny integration tests for anonymous, owner, opponent, follower, blocked, moderator, and service contexts.

Action API contract

```
client_action = game_id + seq + action + payload + expected_state_hash +
                idempotency_key
```

1. Authenticate user or anonymous signed session and load current authoritative game state.
2. Verify game is active, action sequence is next, expected state hash matches, and idempotency key has not produced another result.
3. Validate offered card, role eligibility, tool availability, timer, queue rules, and action payload.
4. Append the action, derive the next state deterministically, persist new hash in one transaction, and return the authoritative response.
5. At completion, enqueue score/receipt/leaderboard/rating settlement jobs with deduplication keys.
6. Log integrity signals separately from product analytics; never trust client-computed score as official.

Realtime events

Channel/event	Payload rule	Consumers
game:{id}:state	Authoritative state version and allowed public delta; never leak future offers.	Active client, reconnecting client.
match:{id}:settled	Outcome, score summaries, rating-event IDs, rematch action.	Both players, notification service.
user:{id}:progress	XP/streak/achievement/rating summary delta.	Owner only.
leaderboard:{scope}:delta	Bounded rank changes or snapshot-ready signal.	Visible leaderboard clients.
creator_event:{id}:stats	Participant count, settled aggregate selections after spoiler threshold.	Creator/admin and public event page.

Table 46. Realtime channel boundaries and leak prevention.

Background jobs

Job	Trigger	Idempotency / retry behavior
score_game	Draft completion.	Keyed by game + ruleset; deterministic output; dead-letter after bounded retries.
settle_match	Both valid results available.	Unique match settlement; transaction creates rating ledger events.
update_daily_percentiles	Incremental results and board closure.	Periodic approximate during day; exact immutable snapshot at close.
refresh_leaderboards	Rating/season event batches.	Materialized snapshot with version/time.
generate_share_asset	Completed result or explicit share.	Keyed by result + template version; store reusable asset.
send_notification	Outbox record.	At-least-once delivery with provider dedupe and user preference check.
publish_release	Approved model/data release.	Multi-stage, resumable, checksummed, manual promotion gate.
integrity_review	Signals exceed threshold or top-ranked result.	No automatic punishment; queue evidence bundle for review.

Table 47. Durable job system and idempotency requirements.

Version compatibility

- ❑ Every peak window references exactly one model version and one input data release.
- ❑ Every board references model, ruleset, card pool, and lineup-model versions.
- ❑ Every result stores a snapshot of selected cards and receipt, not only foreign keys to mutable display records.
- ❑ Leaderboards disclose which versions are comparable and split records when scoring changes materially.
- ❑ The client refuses to submit actions under a ruleset it cannot interpret and refreshes safely.
- ❑ Release tooling includes a compatibility matrix and migration/rollback plan.
- ❑ Historic public links continue to resolve after updates.

PART 14

Security, anti-cheat, moderation, privacy, and legal

A ranked game is a trust product. Integrity, safety, and rights strategy must be designed before scale exposes the weaknesses.

Threat model

Threat	Example	Primary defenses
Client manipulation	Submitting a card not offered, changing role eligibility, altering score payload.	Server-authoritative state/actions; signed state hashes; ignore client score.
Future-offer leakage	Inspecting network/cache to see later cards before deciding.	Send only current legal offers; encrypting a full tree is not a substitute.
Board solving	Automated agents compute optimal choices for ranked play.	Private active seeds, timing/behavior signals, challenge separation, top-result review, adaptive board pool.
Multi-account boosting	Users repeatedly match controlled accounts or intentionally lose.	Pairing caps, device/network signals with privacy limits, graph analysis, minimum unique opponents.
Bot farming	Automated completion for XP, streaks, or event standings.	Rate limits, Turnstile/risk challenges where appropriate, action-timing models, XP caps.
API scraping / denial	High-volume requests or public dataset abuse.	CDN/cache, quotas, keys for bulk API, WAF, public downloads for legitimate large access.
Account takeover	Credential stuffing, stolen session, malicious OAuth flow.	Provider security, MFA option, session/device management, anomaly alerts, secure cookies.
Injection / XSS	Display names, creator titles, share text, query state.	Validation, escaping, CSP, safe Markdown policy, parameterized SQL.
Leaderboard harassment	Offensive handles, impersonation, targeted challenge spam.	Name rules, reporting, blocks, rate limits, moderation tools, verified creator process.
Admin abuse / mistakes	Untracked score or moderation changes.	Least privilege, audit log, dual approval for critical actions, reversible ledgers.

Table 48. Launch threat model and defenses.

Anti-cheat principles

- **Do not punish from one weak signal.** Use evidence bundles, thresholds, human review for high-impact actions, and an appeal process.
- **Protect the board, not the formula.** The model can remain transparent; ranked integrity depends on hidden future offers, authoritative actions, and fair comparisons.
- **Separate casual and ranked.** Public seeds, solver experimentation, and custom tools are welcome in Practice; ranked boards carry stronger controls.
- **Make optimality hard to exploit.** Large candidate pools, controlled diversity, hidden branches, and lineup interactions reduce simple memorization while keeping the model explainable.
- **Watch the top.** Automatically integrity-review new global records, Legend promotions, tournament qualification, and anomalous win streaks.
- **Record enough to replay.** Store action sequence, server timing, state hashes, client version, and bounded integrity metadata with retention/privacy rules.

Account and session security

- Use secure, HttpOnly, SameSite cookies and strict redirect allowlists for authentication.

- ☐ Support passkeys or MFA as the platform matures; require stronger authentication for admins and high-value creator accounts.
- ☐ Expose active sessions/devices and allow revocation.
- ☐ Rate-limit login, handle lookup, challenge creation, follows, reports, and all state-changing APIs.
- ☐ Use CSRF defenses, origin validation, CSP, secure headers, dependency scanning, and secret rotation.
- ☐ Keep service-role credentials server-only; no privileged database key in browser bundles.
- ☐ Use Turnstile or equivalent selectively for suspicious/signup flows, not on every normal game action.
- ☐ Maintain an incident response runbook, security contact, and vulnerability disclosure path.

Moderation scope

Surface	Launch policy
Display name / bio	Allow limited text; profanity/impersonation checks; report, forced rename, and appeal.
Creator event title/description	Sanitize and moderate; verified badge is controlled, revocable, and never purchasable as authenticity.
Reactions	Use a small safe set; no free-form match chat at launch.
Profile image	Use generated avatars or tightly controlled upload pipeline; defer arbitrary uploads if not necessary.
Challenge spam	Per-user limits, accept-only settings, block list, and follow-based permissions.
Public leaderboard	Hide suspended names immediately while preserving competition audit internally.
Reports	Categories for harassment, impersonation, cheating, offensive content, privacy, and IP; status/appeal process.

Table 49. Moderation scope keeps the initial community feature set manageable.

Privacy and data minimization

- **Collect only what the feature requires.** Auth identity, handle, timezone, broad optional region, game history, progress, and settings are sufficient for launch.
- **Do not expose precise location.** Regional leaderboards use broad self-selected regions with minimum population thresholds.
- **Separate product analytics and integrity data.** Document purposes, access, retention, and deletion behavior.
- **Consent and control.** Provide privacy settings, data export, account deletion, notification controls, and public-profile visibility.
- **Minor-aware design.** Basketball games attract teenagers. Avoid public DMs/open chat, manipulative monetization, and unnecessary demographic collection.
- **Retention.** Keep immutable competition records where legitimately required, but anonymize/delete personal linkage according to policy and legal review.
- **Vendor review.** List analytics, auth, hosting, email/push, error tracking, and storage vendors in a transparent policy.

Rights and legal review

Statistics and facts can be used differently from protected presentation, photographs, logos, audiovisual content, and database compilations. Player identity and commercial use can also raise publicity and endorsement questions. Source-site terms may restrict automated collection or redistribution even when facts themselves are not protected. This blueprint is product guidance, not legal advice; obtain qualified review before commercial launch.

- ☐ Inventory every data source, collection method, cached artifact, and redistribution field.
- ☐ Review terms of service and license for every source and third-party package.

- ❑ Use original visual design and text-based team/player identification for the initial product.
- ❑ Do not use NBA/team logos, broadcast imagery, card designs, or player photography without permission or a defensible license.
- ❑ Include a clear independent/unofficial disclaimer and avoid implying endorsement.
- ❑ Complete trademark clearance for PEAK3, Arena, Index, Lab, Peak Draft, Triple Crown, and visual marks before major promotion.
- ❑ Create takedown, correction, attribution, and rights-contact procedures.
- ❑ Publish data provenance and avoid redistributing raw source pages or protected editorial content.
- ❑ Review contest/tournament prize rules, age requirements, tax, and jurisdictional restrictions before offering prizes.

Monetization constraints

- **Never sell competitive advantage.** No paid rerolls, stronger tools, easier boards, rating protection, or hidden card intelligence.
- **Good paid value.** Advanced data exports, saved Lab workspaces, creator-event tooling, profile cosmetics, historical season archive cosmetics, and API tiers.
- **Keep public authority public.** Core leaderboards, player pages, methodology, current data release, and formula explanation should remain broadly accessible.
- **Avoid loot-box dynamics.** Collections should be discovery/progression objects, not gambling-like purchases tied to competitive strength.
- **Ads.** Avoid disruptive ads in active games. Sponsorships should be clearly labeled and never alter scoring or ranking.

PART 15

Analytics, experimentation, testing, and operations

Instrument the full choice-to-outcome loop, protect metric integrity, and make every release observable and reversible.

Metric hierarchy

Level	Metric examples	Interpretation
North star	Weekly Meaningful Competitors: unique users completing ≥ 3 valid games and at least one challenge/ranked/social action.	Captures repeat play plus competitive engagement, not empty page views.
Acquisition	Landing visits, playable-share opens, creator-event opens, search referrals.	Where qualified users originate.
Activation	First card selected, first draft completed, first receipt expanded, signup after result, first challenge sent.	Whether the promise becomes real quickly.
Retention	D1/D7/D30, weekly active competitors, daily-board return, streak survival, season return.	Whether habits and goals persist.
Competition	Placement completion, matches/user, rematch rate, rating calibration, queue wait/async completion, upset distribution.	Whether ranked play feels alive and fair.
Research	Player searches, comparison completion, formula node depth, exports, API keys, citation copies.	Whether PEAK3 becomes a reference platform.
Quality	Web Vitals, crash-free sessions, action latency, invalid actions, settlement failures, data issues, support tickets.	Whether execution supports trust.
Health	Reports, blocks, cheating flags, false-positive appeals, notification opt-out, time spent extremes.	Whether growth remains healthy.

Table 50. Product metric hierarchy.

Core event taxonomy

Event	Required properties
landing_viewed	route, referrer class, campaign, device class, auth state, experiment assignments.
mode_selected	mode, queue, source surface, first-session flag.
game_started	game public class, board ID hash, mode, queue, model/ruleset/pool versions.
card_offer_viewed	round, offer archetype distribution, role completion state; do not log hidden future offers to client analytics.
card_selected	round, selected window ID, slot, decision time, alternatives, DNA before/after, tool state.
tool_used	tool, round, state reason category, resulting branch ID.
game_completed	score layers, percentile, efficiency, duration, lineup profile, integrity status.
receipt_expanded	receipt item, depth, entry surface.
challenge_created/opened/completed	public token hash, relationship class, channel, conversion timestamps.
ranked_settled	queue, outcome, rating delta, expected score band, division before/after.

Event	Required properties
formula_node_opened	node ID, model version, trace/compare/custom state, depth.
data_exported	release, format, rows/fields, query class, authenticated plan.

Table 51. Analytics events. Full privacy review and data dictionary are required before collection.

Event quality rules

- Define events in a shared typed package; reject unknown property names in development.
- Attach anonymous/user ID only where necessary and merge anonymous history through an explicit claim flow.
- Attach correlation/game/match IDs for debugging, but hash or omit public-sensitive tokens.
- Never include email, raw OAuth profile, precise location, free-form user text, or secrets in analytics.
- Version event schemas and maintain a changelog; dashboards declare accepted versions.
- Create daily data-quality checks for volume jumps, missing properties, duplicate events, impossible funnels, and timestamp drift.

Experimentation framework

Experiments should improve comprehension, choice quality, and healthy return - not maximize compulsive time. Randomize at user or anonymous-device level, define a primary metric and guardrails before launch, and avoid running overlapping experiments on the same critical interaction without factorial design.

Good experiment	Primary metric	Guardrails
Card information density	Draft completion and decision confidence survey.	Decision time, error rate, mobile overflow, new-user performance.
Receipt order	Receipt expansion and next-game rate.	Dispute/report rate, comprehension quiz, result-screen latency.
Signup timing	Saved-result claim and D7 retention.	First-game completion, abandonment, consent quality.
Daily reminder time	Daily-board return.	Opt-out, notification disable, late-night delivery, complaint rate.
Division progress display	Ranked rematch and season retention.	Anxiety survey, rage quit, support volume.
Formula narrative mode	Node depth and methodology comprehension.	Performance, accessibility, exit rate.

Table 52. Ethical experiment examples.

Testing pyramid

Layer	Required coverage
Model unit/property tests	Formula transforms, weights, missingness, caps, ID stability, window selection, score ranges, lineup bounds, solver invariants.
Schema/contract tests	Python outputs validate against JSON Schema; TypeScript clients generated/tested; API examples run in CI.
Database tests	Migrations up/down strategy, constraints, transactions, RLS allow/deny matrix, rating ledger rebuild.
Service integration	Game action idempotency, state hashes, tool branches, scoring jobs, match settlement, notification outbox.
Component tests	Cards, roles, DNA, tables, charts, formula nodes, focus/keyboard behavior, responsive variants.
E2E journeys	Anonymous game, claim account, ranked placements, challenge, model deep link, export, privacy controls, deletion.

Layer	Required coverage
Visual regression	Key phone/tablet/desktop pages, long names, localization expansion, high contrast, reduced motion.
Accessibility	Automated scans plus manual keyboard/screen-reader/zoom/motion tests.
Load/resilience	Daily reset spike, viral creator board, leaderboard reads, game-action bursts, queue retry, region/provider failure.
Security	Dependency/code scanning, auth/RLS tests, rate limits, injection/CSP, session handling, admin audit.

Table 53. Testing pyramid and launch coverage.

Release gates

- ☐ All model, game-core, contract, and migration tests pass.
- ☐ No unexplained leaderboard changes beyond the reviewed release impact report.
- ☐ Critical user journeys pass on target phone, tablet, desktop, keyboard-only, and at least one screen reader.
- ☐ Performance budgets pass in lab and staging load tests; real-user alert thresholds are configured.
- ☐ Security/RLS regression suite passes; secrets and dependency scans are clean or explicitly waived.
- ☐ Feature flags and rollback plan are verified.
- ☐ Data release checksums, schemas, citations, and changelog are published before or atomically with the web release.
- ☐ Support/moderation/runbook owners know the release-specific changes.
- ☐ Ranked scoring/rules changes occur at announced boundaries unless fixing a critical integrity flaw.

Service objectives

Service	Initial SLO
Public Index/Lab reads	99.9% successful requests monthly; cached pages remain available during partial backend failure.
Game action API	99.95% successful authoritative actions; p95 server response under 300 ms in primary region target.
Match settlement	99% within 60 seconds after both valid results; no duplicate rating events.
Daily board publication	Available before advertised reset; automated fallback board on generation failure.
Leaderboard freshness	Daily boards near-real-time approximate; final snapshot within 5 minutes of close.
Data releases	Checksummed and reproducible; prior release always available.
Notification delivery	Best effort, never blocks game completion or settlement.

Table 54. Initial service-level objectives, to be refined from beta telemetry.

Operational runbooks

- Game actions failing or state-hash mismatch spike.
- Daily board is invalid, leaked, unsolvable, or produces pathological distribution.
- Rating settlement duplicated, delayed, or mathematically incorrect.
- Leaderboard abuse or global-record integrity investigation.
- Model/data release rollback and ranked queue freeze.
- Auth provider outage and anonymous-play fallback.
- Database latency, storage/CDN failure, or queue backlog.

- Offensive impersonating handle or creator-event abuse.
- Data correction, public incident note, and historical result treatment.
- Security incident: session revocation, key rotation, evidence preservation, user notification decision.

PART 16

Roadmap, team plan, and execution cadence

A 16-week focused build can produce a credible public beta if the team protects the core loop and parallelizes model, product, and platform work.

Roadmap overview

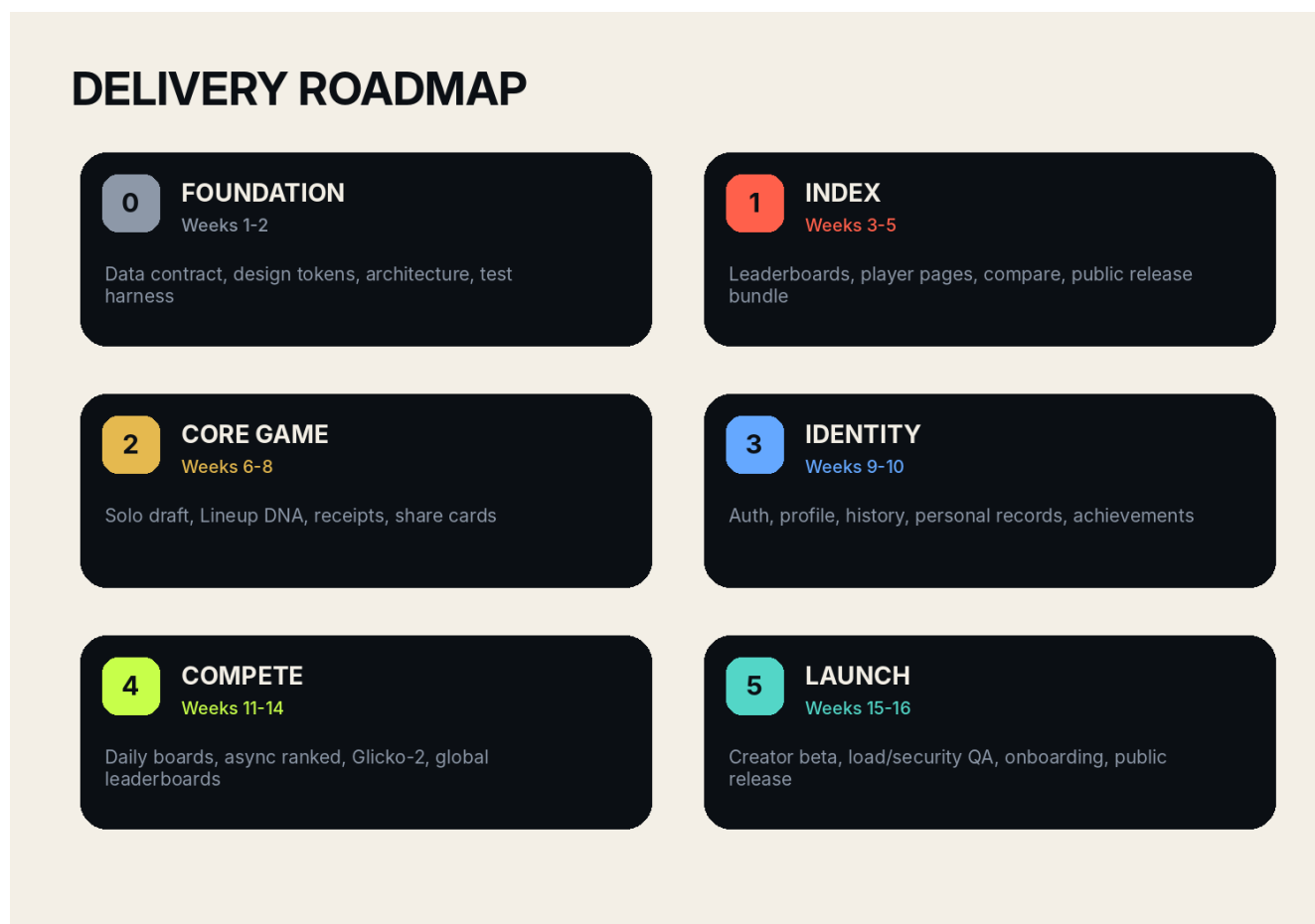


Figure 12. Sixteen-week path from foundations and model contracts to public beta, ranked integrity, and launch hardening.

Phase 0 - alignment and technical discovery (Week 0-1)

Workstream	Deliverables	Exit gate
Product	Final product promise, PO scope, mode rules, vocabulary, key journeys, decision log.	Stakeholders can explain the product and launch scope consistently.
Model	Audit current outputs, authoritative paths, stale helpers, IDs, coverage, release reproducibility.	One documented canonical 1Y/3Y/5Y export path.
Lineup research	Define card profile schema, DNA dimensions, prototype equations, validation corpus.	Experimental status and hypotheses explicitly recorded.
Design	Reference board, tokens, responsive wireframes, card/result/formula prototypes.	Phone core loop approved before high-fidelity expansion.

Workstream	Deliverables	Exit gate
Engineering	Architecture ADRs, monorepo layout, environments, CI, schema strategy, threat model.	Local vertical slice boots and deploys to preview.

Table 55. Phase 0 deliverables.

Phase 1 - vertical slice (Week 2-4)

- Generate one immutable model release with player, season, window, formula graph, and leaderboard artifacts.
- Build global shell, home, 1Y leaderboard, player page skeleton, and formula overview.
- Implement deterministic Practice Peak Draft: five rounds, three offers, roles, Reframe, Hold, local/auth session.
- Implement server-authoritative action state machine and persisted result snapshot.
- Render Lineup DNA and prototype result receipt from a clearly labeled experimental model.
- Instrument critical events and add unit/contract/E2E tests.
- Run 10-20 internal testers through the full loop on mobile and desktop.

VERTICAL-SLICE RULE

At the end of Week 4, a person must be able to land, play a complete exact-peak draft, understand the result, open the selected player/window in the Index, and trace one component in the Lab. No fake buttons or disconnected prototypes.

Phase 2 - product beta core (Week 5-8)

Area	Build
Arena	Daily seeded board, challenge links, spoiler-safe result, match history, practice variants, robust reconnect.
Identity	Auth, handle, profile, anonymous result claim, settings, timezone, privacy.
Progress	XP, levels, basic achievements, personal records, healthy streak with reserve.
Index	Full 1Y/3Y/5Y leaderboards, player Peak Mountain, exact window pages, comparison.
Lab	Four-level formula explorer for all official components, player trace, deep links, release page.
Data	CSV/Parquet/JSON release, dictionary, checksums, examples, API read endpoints.
Quality	RLS tests, accessibility pass, mobile performance budget, visual regression, observability.

Table 56. Phase 2 beta-core build.

Phase 3 - ranked closed beta (Week 9-12)

- Implement async same-board ranked duel and settlement pipeline.
- Integrate Glicko-2 library/implementation with independent 1Y/3Y/5Y queues and append-only rating ledger.
- Add placements, divisions, global/friends/daily leaderboards, and provisional states.
- Build anti-cheat baseline, top-result review console, pairing caps, and abort policy.
- Run board simulations, solver, fairness slice reports, and expert blind lineup reviews.
- Invite 100-500 basketball fans, analysts, and creators; publish clear beta/model caveats.
- Measure rating calibration, queue completion, rematch, confusion, disputes, retention, and device performance.

Phase 4 - launch hardening (Week 13-16)

Area	Launch work
Game balance	Tune board generation, score receipt, role eligibility, lineup bounds, and division thresholds from beta evidence.
Competition	Season zero, ranked archive, leaderboard snapshots, promotion treatment, rating support tools.
Growth	Creator Gauntlet beta, vertical share assets, referral attribution, editorial launch stories.

Area	Launch work
Operations	SLO dashboards, alerts, status page, incident/moderation/security runbooks, support workflow.
Legal	Data/source/brand review, policies, disclaimers, terms, privacy, contest rules if applicable.
Scale	Load test daily reset and viral board; cache/queue tuning; backup/restore drill.
Polish	Microcopy, onboarding, empty/error/loading states, accessibility/manual device pass, localization readiness.

Table 57. Phase 4 public-beta hardening.

Suggested team

Role	Primary ownership	Minimum involvement
Founder / product-model lead	Product truth, PEAK3 methodology, lineup research, prioritization, creator relationships.	Full time.
Product/full-stack engineer	Next.js, game state, APIs, auth, database, deployment.	Full time.
Data/model engineer	Canonical exports, release pipeline, simulations, solver, validation, data provenance.	Full time or strong founder coverage.
Product designer / frontend specialist	Visual system, responsive game, formula interaction, prototyping, accessibility.	Full time through beta or high-intensity contract.
Platform/quality engineer	CI, RLS/security, E2E/load, observability, queues, reliability.	Part time early; increased Week 8 onward.
Basketball research panel	Role tags, historical review, lineup pairwise evaluation, issue triage.	Structured part-time panel.
Legal counsel	Data rights, trademarks, policies, contests, monetization.	Milestone review.
Community/creator lead	Closed beta recruitment, content calendar, moderation/support, creator events.	Part time prelaunch; full/contract near launch.

Table 58. Lean team composition. One person may cover multiple roles, but every ownership area must be explicit.

Execution cadence

- **Weekly planning.** Choose one user-visible outcome, one model/quality objective, and one risk-reduction objective per workstream.
- **Daily integration.** Keep vertical slices deployable; avoid long-lived model/UI branches that diverge.
- **Twice-weekly playtest.** Record phone sessions, decision confusion, score disagreement, and emotional moments.
- **Weekly model council.** Review lineup anomalies, role tags, rank changes, data issues, and experiment proposals with written decisions.
- **Biweekly release.** Ship behind flags, publish internal notes, and run all gates.
- **Monthly public changelog.** Explain meaningful data/model/product changes in plain language.
- **Decision records.** Use short ADRs/product decision records for scoring, rating, roles, privacy, and architecture choices.

Critical path

1. Canonical, versioned PEAK3 exports and stable IDs.
2. Card profile and experimental lineup model with testable bounds.
3. Server-authoritative deterministic game state.
4. Compelling mobile draft and result receipt.
5. Seeded shared boards and challenges.
6. Auth/profile and immutable result history.
7. Ranked settlement and rating ledger.
8. Integrity, accessibility, performance, and operations hardening.

PART 17

Launch, creators, community, and growth

The product should launch as a shared basketball argument people can play, inspect, and recreate.

Positioning

The NBA peak index you can play.

Recommended public shorthand

A longer description: "PEAK3 ranks exact NBA seasons and multi-year peaks with a transparent model. Build a five-player lineup from shared peak boards, compete globally, and inspect exactly why every result happened."

Launch story pillars

Pillar	Story
The rankings	The public can finally explore exact 1Y, 3Y, and 5Y peaks in one consistent, versioned system.
The game	Fans draft exact peaks rather than generic names and compete on the same fair board.
The receipts	Every result exposes talent, coverage, synergy, alternatives, and model provenance.
The openness	Analysts can download the data, reproduce outputs, compare versions, and build their own work.
The ritual	Today's Peak and Triple Crown become recurring global basketball challenges.

Table 59. Launch narrative pillars.

Creator program

1. Invite a small group of basketball YouTubers, TikTok/Instagram creators, podcasters, analytics writers, and team-community hosts into closed beta.
2. Give each creator a stable public profile, event builder, branded challenge landing page, vertical/landscape exports, and audience aggregate report.
3. Create recurring formats: "Beat My Triple Crown," "The Crowd Was Wrong," "Blind Resume," "Peak Budget," and "One Franchise Gauntlet."
4. Do not pay for scripted praise. Ask creators to disagree with the model, investigate receipts, and surface anomalies. Product credibility grows through inspectable debate.
5. Feature community analysis, accepted data corrections, and high-quality alternative interpretations in editorial surfaces.
6. Measure challenge completion and retained players, not view count alone.

Playable sharing

Share type	Contents	Destination
Result card	Lineup, Peak Rating, percentile/division outcome, one strength, one weakness, model version.	Immutable result page with Play This Board.
Challenge card	Challenger handle, mode, spoiler-safe invitation, expiration/rules.	Same hidden board and queue-safe challenge flow.
Comparison graphic	Selected windows, component difference, one conclusion, citation/version.	Interactive comparison page.
Peak Mountain	Career chart, highlighted 1Y/3Y/5Y, canonical URL.	Player page.

Share type	Contents	Destination
Creator board	Event title, creator, participant count, deadline, no leaked offers.	Creator event landing page.
Leaderboard milestone	Division/rank/promotion, season, queue, integrity-verified status.	Profile/leaderboard snapshot.

Table 60. Every share object should lead to a useful interactive destination.

Editorial engine

- **The Crowd Was Wrong.** Analyze the most-passed card that improved outcomes and explain the basketball logic.
- **Peak Disagreements.** Compare public intuition, traditional accolades, performance-only, and full PEAK3 without rage bait.
- **One Decision.** Show the one round that separated top and median Daily Peak lineups.
- **Model Update.** Explain what changed, why, affected rankings, and what remained stable.
- **Peak Atlas.** Explore one era, archetype, franchise, or playoff pattern using Index/Lab visuals.
- **Creator Film Room.** A creator reviews their audience's aggregate choices and strongest unusual lineups.
- **Open Question.** Publish model limitations and invite structured evidence or issue submissions.

Launch sequence

Stage	Audience	Goal
Internal alpha	Team and research panel.	Eliminate state, formula, role, and obvious UX failures.
Friends-and-experts alpha	50-100 serious NBA fans/analysts.	Validate game depth, receipts, terminology, and lineup face validity.
Creator closed beta	100-500 invitees across creator communities.	Stress challenges, share conversion, ranking calibration, moderation.
Public Season Zero	Open access with beta label and archived rules.	Establish daily ritual, global ladder, and operational confidence.
Season One	Polished ranked system, Triple Crown, creator program, stable data release.	Make PEAK3 a recurring destination rather than one viral spike.

Table 61. Controlled launch sequence.

Launch-week programming

1. Day 1: Publish Index, methodology story, and Today's Apex board.
2. Day 2: 3Y Prime board and creator "Who peaked higher?" collaborations.
3. Day 3: 5Y Foundation board and open data release examples.
4. Day 4: First global Triple Crown; live aggregate dashboard after spoiler close.
5. Day 5: The Crowd Was Wrong analysis and public formula-explorer walkthrough.
6. Weekend: Creator Gauntlets, ranked placement race, community issue/feedback session.
7. Week 2: Publish launch data, anomalies, fixes, and next-season schedule to demonstrate transparency.

Community governance

- Publish an issue taxonomy: data correction, role eligibility, methodology, UI, game balance, accessibility, security.
- Separate empirical data corrections from normative scoring disagreements.
- Require reproducible evidence for model/data issues; make accepted changes visible in the changelog.
- Create a public model council note for major scoring decisions, including dissent and uncertainty.
- Credit substantive community contributors where appropriate.
- Keep ranked hotfix authority centralized and auditable; do not let polls directly rewrite formulas.

PART 18

Scope priorities, acceptance criteria, and open decisions

The project succeeds by shipping one complete, trusted loop before expanding into every possible basketball game.

P0 - public beta must contain

Area	P0 scope
Core brand/shell	PEAK3 umbrella, Index/Arena/Lab navigation, responsive shell, search, model-version status.
Arena	Practice, Daily Peak, direct challenges, async ranked duels; 1Y/3Y/5Y; five roles; Reframe/Hold; DNA; receipts.
Ranked	Glicko-2 queue ratings, placements, divisions, global/daily/friends leaderboards, immutable rating ledger.
Progress	Account/profile, result history, XP/level, personal records, modest achievements, healthy streak.
Index	1Y/3Y/5Y leaderboards, player page/Peak Mountain, exact window page, comparison.
Lab	Interactive official formula overview through metric transformation, player trace, release/download pages.
Data	Versioned CSV/Parquet/JSON, schema/dictionary/provenance/checksums, public read API.
Trust	Server authority, version snapshots, anti-cheat baseline, RLS, moderation/report/block, privacy controls, legal review.
Quality	Mobile-first polished UI, accessibility, Core Web Vitals targets, observability, E2E/visual/load/security gates.

Table 62. P0 public-beta scope.

P1 - build after the core proves retention

Area	P1 scope
Competition	Triple Crown rating/events, creator Gauntlets, constraint drafts, seasonal points and tournaments, rivals.
Game discovery	Blind Resume Sprint, Who Peaked Higher, curated franchise/era boards, richer achievements/collections.
Lab	Full node provenance, custom-weight sandbox, model-version compare, saved data workspaces, richer exports.
Creator	Event builder, audience decision analytics, templates, verified profiles, scheduled events.
Platform	Passkeys/MFA, expanded notifications, developer keys/quotas, advanced moderation/integrity console.

Table 63. P1 expansion scope.

P2 - explicitly defer

- Live synchronous rooms, spectators, chat, brackets, and large tournaments.
- Mirror Draft and Peak Auction with realtime bidding/selection.
- Era Court simulation or possession-level matchup engine.
- Native iOS/Android applications before mobile web proves product-market fit.
- Open marketplace, tradable cards, paid randomized collectibles, or token systems.

- Broad user-generated formulas that can masquerade as official PEAK3.
- Expansion to WNBA, college, international leagues, or other sports before NBA operations are stable.
- Complex social feed or direct messaging.

P0 acceptance criteria

First-time experience

- ❑ At least 80% of moderated new-user testers can explain the objective after the intro and make the first choice without assistance.
- ❑ A visitor can complete an anonymous mobile game, understand the result, and create a challenge in under five minutes.
- ❑ Sign-up after result preserves the game, challenge, and progress without duplicate identity or lost state.

Game and scoring

- ❑ All legal action paths replay deterministically from the stored board/rules/action sequence.
- ❑ No client modification can produce an official card, role, tool, score, or rating event not accepted by the server.
- ❑ Every result exposes talent, coverage, synergy, model/ruleset versions, and at least three reproducible receipts.
- ❑ Board generator passes feasibility, diversity, distribution, and solver checks before official publication.
- ❑ Lineup model passes documented bounds, invariant, adversarial, slice, and expert-review gates.

Ranked

- ❑ Placements, provisional status, Glicko-2 update, draws, inactivity uncertainty, and corrections are covered by deterministic tests.
- ❑ Ratings can be rebuilt exactly from the ledger.
- ❑ Global rank excludes ineligible/inactive/suspended accounts according to published rules.
- ❑ Same-board competitive equivalence is verifiable; future offers do not reach clients early.
- ❑ Top results and promotions produce integrity review signals without blocking normal play unnecessarily.

Index and Lab

- ❑ A player/window score shown in any UI matches the immutable release row and selected model version.
- ❑ Every formula contribution can be traced from final score to raw input and provenance metadata.
- ❑ Leaderboards, comparisons, charts, downloads, and API examples use stable IDs and disclose version/completeness.
- ❑ Old release pages and game links remain functional after a new release.

Quality

- ❑ No critical accessibility issues in automated and manual launch audit.
- ❑ Target Core Web Vitals are met at the 75th percentile in beta traffic or have an approved blocker plan.
- ❑ Daily reset and projected creator-spike load tests meet action, settlement, and read SLOs.
- ❑ Incident, rollback, moderation, data correction, and rating correction drills have been executed.
- ❑ Privacy/terms/disclaimer/data-source/trademark review is complete before broad public promotion.

Decisions to make in Week 0

Decision	Recommended default	Evidence needed to change it
Product naming	PEAK3 / INDEX / ARENA / LAB; Peak Draft; Triple Crown.	Trademark review or strong user confusion evidence.
Launch roles	Five functional roles defined in Part 5.	Testing shows rigid positions are clearer without historical distortion.
Launch tools	One Reframe and one Hold.	Playtests show insufficient agency or one tool dominates.
Lineup weight hypothesis	78% talent / 14% coverage / 8% synergy.	Validation and optimization evidence; do not tune to anecdotes only.
Rank algorithm	Glicko-2, independent 1Y/3Y/5Y.	Simulation/calibration indicates another published system materially improves outcomes.
Division names	Prospect → Rotation → Starter → All-Star → All-NBA → MVP → Legend.	Brand testing, sensitivity, or localization concerns.
Season length	10 weeks as an initial planning assumption.	Beta cadence, tournament schedule, and retention data.
Anonymous play	Full first game and share; sign in to preserve/rank.	Abuse makes a narrower anonymous flow necessary.
Frontend	Next.js/TypeScript with D3/Plot.	Team expertise or measured platform constraints.
Backend/data	Supabase/Postgres plus server APIs/jobs; Python model artifacts.	Scale or compliance requirement unsupported by stack.

Table 64. High-impact Week 0 decisions and evidence standards.

Immediate implementation backlog - first 30 issues

1. Create product decision record for P0 scope, terminology, and non-goals.
2. Audit canonical PEAK3 1Y/3Y/5Y code paths and remove/flag stale helpers.
3. Define stable IDs for player, season, player-season, window, model release, and methodology node.
4. Design versioned release manifest and player_seasons/peak_windows schemas.
5. Generate first web-ready immutable release and checksum validation.
6. Define card-profile and role-eligibility schemas with authority/provenance fields.
7. Build Lineup DNA research notebook and synthetic invariant suite.
8. Create curated lineup pairwise validation dataset and reviewer workflow.
9. Implement monorepo, CI, preview deployment, lint/type/test gates.
10. Create design tokens and responsive shell.
11. Build Peak Card component states and Storybook/test matrix.
12. Build phone-first five-round Practice Draft prototype.
13. Implement deterministic board seed and offer-tree generator.
14. Implement server-authoritative action/state hash API.
15. Implement Reframe/Hold branches and replay tests.
16. Implement lineup result/receipt interface against placeholder versioned scorer.
17. Build 1Y leaderboard and exact window page from generated release.
18. Build player search and initial Peak Mountain.
19. Create methodology JSON graph and formula overview.
20. Implement player trace for one component end to end.
21. Configure Supabase Auth, profiles, RLS, and anonymous-claim flow.
22. Persist game/action/result snapshots and history.
23. Implement share/challenge public token and spoiler-safe metadata.

- 24. Create typed analytics event package and first funnel dashboard.
- 25. Add Playwright anonymous game and claim-account journeys.
- 26. Add accessibility and visual-regression CI for game/result/index/formula.
- 27. Implement public data release page and CSV/Parquet downloads.
- 28. Prototype Glicko-2 simulation and rating ledger schema.
- 29. Write threat model, anti-cheat baseline, and moderation scope.
- 30. Recruit first 20 structured playtesters and publish a feedback rubric.

Final product test

Can a new fan make a satisfying choice immediately, can an expert inspect every assumption, and can two players compete fairly on the same evidence?

The standard for every major PEAK3 decision

PART A

Appendices: implementation reference

Schemas, event contracts, launch checklists, and curated internet resources for the team.

Appendix A1 - game configuration example

Key	Illustrative value
mode	apex_1y
rounds	5
offers_per_round	3
roles	lead_creator, guard_wing, wing_forward, forward_big, anchor
tools	reframe:1; hold:1
model_version	peak3@1.x
lineup_model_version	lineup@0.x-experimental
ruleset_version	peak-draft@1.0.0
timer	none (daily/practice); configured for ranked tier
ranked_result_key	official_lineup_score; draw tie-breaks documented
share_visibility	spoiler_safe until completed/settled

Table 65. Configuration should be data-driven, versioned, signed, and validated.

Appendix A2 - methodology graph node

Field	Meaning
id	Stable semantic identifier, e.g. statistical_impact.bpm.standardize.
model_version	Exact model version in which the node is valid.
label / short_label	Human-readable names for canvas and breadcrumb.
node_type	model, component, metric, transform, source, output.
description_plain	Short accessible explanation.
equation	Formal expression in a renderable notation.
inputs / outputs	IDs, fields, units, ranges, and dependency edges.
weight / cap	Official values or references to parameter registry.
missing_policy	Renormalization, exclusion, imputation, flag, or not applicable.
coverage	Seasons/players for which the input is available.
limitations	Known interpretive or data limits.
examples	Selected player/window trace payload IDs.
sources	Provenance record IDs and external citation metadata.
tests	Invariant/regression test IDs and public description.
change_log	Previous node/version and reason for change.

Table 66. Machine-readable Formula Explorer node schema.

Appendix A3 - peak window public fields

Field group	Representative fields
Identity	window_id, player_id, player_name, duration_years, season_start, season_end, seasons, team_ids.
Version	model_version, input_release, window_algorithm_version, published_at.
Scores	prime_raw, prime_score, performance_only, regular_performance, component scores, teammate adjustment.
Rank	global rank by duration/score type, era rank, position/role rank, percentile.
Window mechanics	season weights, completeness treatment, gap status, eligibility flags.
Context	awards summary, postseason summary, team success, role/workload tags.
Quality	missing metric flags, source coverage, confidence/completeness category, correction status.
Citation	canonical URL, release URL, methodology URL, row checksum.

Table 67. Suggested public peak-window schema.

Appendix A4 - rating update safeguards

- ☐ Use a tested Glicko-2 implementation and verify against published example calculations.
- ☐ Store rating, rating deviation, volatility, rating period, and algorithm version in every event.
- ☐ Treat a draw correctly; do not convert close scores into arbitrary win/loss unless the rules say so.
- ☐ Update both participants in one settlement transaction or atomically publish compensating events.
- ☐ Prevent duplicate match settlement through unique keys and idempotent jobs.
- ☐ Cap or investigate impossible/extreme deltas caused by corrupt input, not by secretly altering the algorithm.
- ☐ Simulate new-player placements, inactive return, repeated opponents, upset rates, and division distribution before launch.
- ☐ Publish high-level rating rules and division eligibility; keep anti-abuse thresholds private.
- ☐ Build an exact ledger replay and admin correction path before the first public ranked match.

Appendix A5 - board generation acceptance checks

Check	Fail condition
Stable generation	Same seed + versions does not reproduce the same offer tree.
Role feasibility	No legal path can fill all roles without a tool, unless the mode explicitly allows it.
Tool validity	Reframe/Hold branches contain invalid cards, duplicate illegal windows, or dead state.
Difficulty distribution	Simulations show nearly every policy receives the same result or optimal choices are trivial.
Diversity	Player/era/team/archetype concentration exceeds configured caps without a themed-board reason.
Score health	Board optimum/floor/spread falls outside launch bounds or exploits a known lineup rule.
Leak safety	Client payload contains future offers or seed material before permitted.
Version completeness	Any card lacks model, profile, role, completeness, or source snapshot required by rules.

Check	Fail condition
Human review	Featured/ranked board generates obviously nonsensical choices or misleading historical presentation.

Table 68. Automated and human checks before official board publication.

Appendix A6 - analytics definitions

Metric	Definition
First-game completion	Anonymous/auth sessions that start a game and produce a valid result within 30 minutes.
Activation	First-game completion plus at least one receipt expansion, challenge action, saved comparison, or ranked-placement start.
D1 / D7 retention	Activated users with a qualifying active event on the corresponding local-day window; publish exact convention.
Weekly Meaningful Competitor	Unique user with ≥ 3 valid completed games and ≥ 1 ranked/challenge/social competitive action in 7 days.
Playable share conversion	Unique recipients who open a share and start the linked board / unique recipients who open.
Challenge completion	Received challenges completed before expiration / valid challenge opens.
Ranked rematch rate	Settled ranked matches followed by another ranked match within the configured time window.
Receipt engagement	Completed games with at least one detailed receipt or decision replay expansion.
Formula depth	Maximum unique methodology level reached in a session, not raw click count.
Healthy session guardrail	Distribution of session length and repeated games, plus self-reported enjoyment/frustration and notification opt-out.

Table 69. Canonical product metric definitions should live in code and warehouse documentation.

Appendix A7 - design QA matrix

Surface	States to capture
Peak card	Default, focus, hover, selected, disabled, hidden identity, missing data, long name, 1Y/3Y/5Y, high contrast.
Draft board	Round 1/5 and 5/5, Hold active, Reframe confirmation, role conflict, reconnect, timeout, narrow phone, landscape.
Result	Personal best, promotion, loss, draw, provisional, receipt expanded, long lineup names, share sheet.
Leaderboard	Anonymous, unranked, provisional, ranked, tied rank, suspended hidden, empty friends, huge rank, pagination.
Peak Mountain	Short/long career, gaps, tied peaks, compare overlay, reduced motion, table fallback.
Formula Explorer	All four levels, keyboard focus, selected trace, compare, custom unofficial, missing source, loading/error.
Profile	Private/public/followers views, blocked user, no history, multiple queue ratings, achievements.
Data Lab	No results, large result, mobile table, chart controls, export pending/error, saved query.

Table 70. Visual regression and design QA matrix.

Appendix A8 - launch readiness checklist

- ☐ Product promise and rules are consistent across home, onboarding, help, result, and marketing.
- ☐ P0 mode matrix and ranked eligibility are frozen for Season Zero.
- ☐ Current model/data/ruleset versions are visible and linked.
- ☐ First daily boards have been generated, simulated, manually reviewed, and scheduled with fallbacks.
- ☐ Ranked rating simulation and ledger rebuild are signed off.
- ☐ Creator/event load test exceeds expected launch spike.
- ☐ Core Web Vitals and action latency dashboards are live.
- ☐ Moderation, reports, blocks, handle enforcement, and creator verification work.
- ☐ Terms, privacy, disclaimer, data sources, accessibility statement, and status/support pages are live.
- ☐ Data release, dictionary, provenance, examples, checksums, and changelog are downloadable.
- ☐ Incident, rollback, database restore, rating correction, and compromised-board drills are complete.
- ☐ Launch creators have instructions that encourage honest debate and disclose beta limitations.
- ☐ Feedback/issue intake routes to clear owners and publicly visible change communication.
- ☐ Season Zero has a published schedule, end condition, awards, and archival plan.

Curated internet resources

These references are selected for concrete implementation questions. They are not aesthetic templates to copy. Capture screenshots and notes into an internal reference board, then translate principles into the PEAK3 system.

1. **PEAK3 repository** - Current engine, methodology, CLI, data pipeline, tests, and canonical outputs. [Open resource](#)
2. **82-0 official rules** - Study the one-sentence loop, rerolls, share format, and frictionless onboarding. [Open resource](#)
3. **The Verge: 82-0 phenomenon** - Context on why a minimal stats game became culturally sticky and easy to discuss. [Open resource](#)
4. **SixRings official guide** - Study exact-season drafting, tools, lineup bonuses, and mode depth. [Open resource](#)
5. **GeoGuessr ranked guide** - Reference divisions, competitive queues, and readable ranked progression. [Open resource](#)
6. **Glicko-2 official site/paper** - Implement and verify rating, rating deviation, volatility, and rating periods. [Open resource](#)
7. **Chess.com leaderboards** - Reference how global/friends categories and leaderboards are presented. [Open resource](#)
8. **Chess.com streaks** - Reference streak definition and user-facing progression language. [Open resource](#)
9. **Chess.com achievements** - Reference discoverable and hidden achievement systems. [Open resource](#)
10. **Strava segments** - Reference persistent shared challenges, personal records, and social comparison. [Open resource](#)
11. **Strava leaderboard filters** - Reference global/following/age/weight-style filtered competition while protecting privacy. [Open resource](#)
12. **Duolingo streak research** - Reference habit formation, streak protection, and less punitive recurrence. [Open resource](#)
13. **D3** - Custom data visualization, scales, transitions, interaction, and SVG/canvas composition. [Open resource](#)
14. **D3 hierarchy** - Formula Explorer hierarchy, zoomable node structures, and tree/partition layouts. [Open resource](#)
15. **Observable Plot** - Concise, accessible statistical charts for the Index and Lab. [Open resource](#)
16. **The Pudding resources** - Study editorial data storytelling, scrollytelling restraint, and visual explanation. [Open resource](#)
17. **Apple Human Interface Guidelines** - Reference hierarchy, feedback, motion purpose, accessibility, and platform behavior. [Open resource](#)
18. **Material Design** - Reference component states, layout, feedback, and interaction patterns without copying the visual brand. [Open resource](#)
19. **web.dev Core Web Vitals** - Set real-user performance targets and understand LCP, INP, and CLS. [Open resource](#)
20. **W3C reduced motion technique** - Implement and test prefers-reduced-motion behavior. [Open resource](#)

- 21. Next.js production checklist** - Review production rendering, security, caching, metadata, bundle, and deployment considerations. [Open resource](#)
- 22. Supabase Auth for Next.js** - Implement modern server/client auth flows and session handling. [Open resource](#)
- 23. Supabase Row Level Security** - Design and test user-data policies. [Open resource](#)
- 24. Supabase Realtime authorization** - Protect per-user and per-match realtime channels. [Open resource](#)
- 25. PostgreSQL row security** - Understand the database-level behavior and policy semantics. [Open resource](#)
- 26. Cloudflare Turnstile** - Use selective bot challenges in signup/suspicious flows. [Open resource](#)
- 27. Playwright visual comparisons** - Build screenshot regression for critical responsive states. [Open resource](#)
- 28. Playwright accessibility testing** - Integrate automated accessibility checks while retaining manual testing. [Open resource](#)
- 29. OpenTelemetry JavaScript** - Trace web/API/jobs and propagate game/match correlation IDs. [Open resource](#)
- 30. Vercel Speed Insights** - Collect real-user performance signals for a Next.js deployment. [Open resource](#)
- 31. U.S. Copyright Office FAQ** - Understand the distinction between facts and protected expression/compilations. [Open resource](#)
- 32. NBA Terms of Use** - Review the league site's asserted rights and content-use restrictions. [Open resource](#)
- 33. USPTO trademark basics** - Begin brand-name and mark clearance before major public investment. [Open resource](#)

Source and assumption note

Product recommendations in this document synthesize the current PEAK3 repository, public descriptions of adjacent games, established ranked-rating systems, interface/design guidance, and implementation documentation available as of June 27, 2026. Specific lineup weights, XP values, division thresholds, season length, performance budgets, and service targets are starting hypotheses or planning targets unless explicitly labeled canonical. They must be tested, reviewed, and versioned before becoming official game rules.

Closing direction

Build the most explainable basketball drafting game, not the busiest one.

Implementation north star

The strongest first release is a complete loop: exact peaks, fair shared boards, a beautiful five-round draft, live Lineup DNA, transparent receipts, a global ranked ladder, and a public Index/Lab that makes every number inspectable. When that loop becomes a habit, the later modes have a foundation. Without that foundation, additional modes only multiply confusion.